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(54) **MULTI-STAGE PROCESS AND DEVICE UTILIZING STRUCTURED CATALYST BEDS AND REACTIVE DISTILLATION FOR THE PRODUCTION OF A LOW SULFUR HEAVY MARINE FUEL OIL**

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(58) **Field of Classification Search**
None
See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

3,163,593 A	12/1964	Webster et al.
3,227,645 A	1/1966	Frumkin et al.

(Continued)

FOREIGN PATENT DOCUMENTS

CA	1054130	5/1979
CA	1060370	8/1979

(Continued)

OTHER PUBLICATIONS

A. M. Aitani, M.F. Ali, H.H. Al-Ali, "A Review of Non-Conventional Methods for the Desulfurization of Residual Fuel Oil", Petroleum Science and Technology, 2000 18:5-6, 537-553, Marcel Dekker Inc. New York, New York USA.

(Continued)

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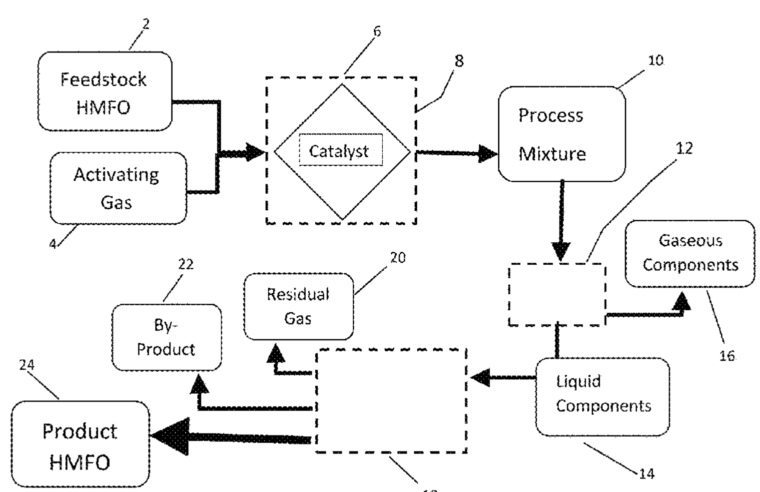
(57) **ABSTRACT**

A multi-stage process for the production of an ISO8217 compliant Product Heavy Marine Fuel Oil from ISO 8217 compliant Feedstock Heavy Marine Fuel Oil involving a core process under reactive conditions in a Reaction System composed of one or more reaction vessels, wherein one or more of the reaction vessels contains one or more catalysts in the form of a structured catalyst bed and is operated under reactive distillation conditions. The Product Heavy Marine Fuel Oil has a sulfur level has a maximum sulfur content (ISO 14596 or ISO 8754) between the range of 0.05 mass % to 1.0 mass. A process plant for conducting the process for conducting the process is disclosed.

13 Claims, 6 Drawing Sheets



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(56) **References Cited**

U.S. PATENT DOCUMENTS

3,234,121	A		2/1966	Maclaren
3,287,254	A		11/1966	Paterson
3,306,845	A		2/1967	Poll
3,531,398	A		9/1970	Adams
3,544,452	A		12/1970	Jaffe
3,551,328	A		12/1970	Cole et al.
3,562,800	A		2/1971	Carlson et al.
3,577,353	A		5/1971	White
3,658,681	A		4/1972	Wilson et al.
3,668,116	A		6/1972	Adams et al.
3,684,688	A		8/1972	Roselius
3,694,343	A	*	9/1972	Christensen C10G 7/00 208/100
3,749,664	A		7/1973	Mickleson
3,809,644	A		5/1974	Johonson et al.
3,814,683	A		6/1974	Christman et al.
3,893,909	A		7/1975	Selvidge
3,902,991	A		9/1975	Christensen et al.
3,910,834	A		10/1975	Anderson
3,968,026	A		7/1976	Frayer et al.
4,006,076	A		2/1977	Christensen et al.
4,017,382	A		4/1977	Bonnell et al.
4,051,021	A		9/1977	Hamner
4,054,508	A		10/1977	Milstein et al.
4,089,774	A		5/1978	Oleck et al.
4,115,248	A		9/1978	Mulaskey
4,118,310	A		10/1978	Frayer et al.
4,138,227	A		2/1979	Wilson et al.
4,225,421	A		9/1980	Hensley, Jr. et al.
4,267,033	A		5/1981	Heck et al.
4,306,964	A		12/1981	Angevine
4,357,263	A		11/1982	Heck et al.
4,404,097	A		9/1983	Angevine et al.
4,420,388	A		12/1983	Bertolacini et al.
4,430,198	A		2/1984	Heck et al.
4,460,707	A		7/1984	Simpson
4,498,972	A		2/1985	Toulhoat et al.
4,499,203	A		2/1985	Toulhoat et al.
4,510,042	A		4/1985	Billon et al.
4,548,710	A		10/1985	Simpson
4,552,650	A		11/1985	Toulhoat et al.
4,604,185	A		8/1986	Mc Conaghy, Jr. et al.
4,645,584	A		2/1987	Didchenko et al.
4,925,554	A		5/1990	Sato et al.
5,167,796	A		12/1992	Didchenko et al.
5,306,419	A		4/1994	Harrison et al.
5,342,507	A		8/1994	Dai et al.
5,374,350	A		12/1994	Heck et al.
5,389,595	A		2/1995	Simpson et al.
5,391,304	A		2/1995	Lantos
5,401,392	A		3/1995	Courty et al.
5,417,846	A		5/1995	Renard
5,543,036	A		8/1996	Chang et al.
5,591,325	A		1/1997	Higashi
5,622,616	A		4/1997	Porter et al.
5,686,375	A		11/1997	Iyer et al.
5,759,385	A		6/1998	Aussillous et al.
5,779,992	A		7/1998	Higashi
5,868,923	A		2/1999	Porter et al.
5,882,364	A		3/1999	Dilworth
5,888,379	A		3/1999	Ushio et al.
5,897,768	A		4/1999	McVicker et al.
5,917,101	A		6/1999	Munoz
5,922,189	A		7/1999	Santos

(56) **References Cited**

U.S. PATENT DOCUMENTS

5,928,501 A 7/1999 Sudhakar et al.
5,948,239 A 9/1999 Virdi et al.
5,958,816 A 9/1999 Neuman et al.
5,961,709 A 10/1999 Hayner et al.
5,976,361 A 11/1999 Hood et al.
5,997,723 A 12/1999 Wiehe et al.
6,017,443 A 1/2000 Buchanan
6,117,306 A 9/2000 Morel et al.
6,136,179 A 10/2000 Sherwood, Jr. et al.
6,160,193 A 12/2000 Gore
6,162,350 A 12/2000 Soled et al.
6,171,477 B1 1/2001 Morel et al.
6,171,478 B1 1/2001 Cabrera et al.
6,193,766 B1 2/2001 Jordan
6,203,695 B1 3/2001 Harle et al.
6,207,041 B1 3/2001 Morel et al.
6,217,749 B1 4/2001 Espeillac et al.
6,251,262 B1 6/2001 Hatanaka et al.
6,251,263 B1 6/2001 Hatanaka et al.
6,265,629 B1 7/2001 Fava et al.
6,299,759 B1 10/2001 Bradway et al.
6,303,531 B1 10/2001 Lussier et al.
6,306,287 B1 10/2001 Billon et al.
6,306,289 B1 10/2001 Hayashi et al.
6,328,880 B1 12/2001 Yoshita et al.
6,344,136 B1 2/2002 Butler et al.
6,383,975 B1 5/2002 Rocha et al.
6,402,940 B1 6/2002 Rappas
6,406,615 B1 6/2002 Iwamoto et al.
6,514,403 B1 * 2/2003 Louie C10G 65/12
208/89
6,540,904 B1 4/2003 Gun et al.
6,554,994 B1 4/2003 Reynolds et al.
6,566,296 B2 5/2003 Plantenga et al.
6,576,584 B1 6/2003 Iijima et al.
6,589,908 B1 7/2003 Ginestra et al.
6,620,313 B1 9/2003 Soled et al.
6,649,042 B2 11/2003 Dassori et al.
6,656,348 B2 12/2003 Dassori et al.
6,656,349 B1 12/2003 Fujita et al.
6,712,955 B1 3/2004 Soled et al.
6,733,659 B1 5/2004 Kure et al.
6,774,275 B2 8/2004 Smith, Jr. et al.
6,783,661 B1 8/2004 Briot et al.
6,797,153 B1 9/2004 Fukuyama et al.
6,858,132 B2 2/2005 Kumagai et al.
6,860,987 B2 3/2005 Plantenga et al.
6,863,803 B1 3/2005 Riley et al.
6,929,738 B1 8/2005 Riley et al.
6,984,310 B2 1/2006 Ginstra et al.
7,001,503 B1 2/2006 Koyama et al.
7,108,779 B1 9/2006 Thakkar
7,119,045 B2 10/2006 Magna et al.
7,166,209 B2 1/2007 Dassori
7,169,294 B2 1/2007 Abe et al.
7,232,515 B1 6/2007 Demmin et al.
7,244,350 B2 7/2007 Martin et al.
7,265,075 B2 9/2007 Tsukada et al.
7,288,182 B1 10/2007 Soled et al.
7,384,537 B2 6/2008 Nagamatsu et al.
7,402,547 B2 7/2008 Wellington et al.
7,413,646 B2 8/2008 Wellington et al.
7,416,653 B2 8/2008 Wellington
7,449,102 B2 11/2008 Kalnes
7,491,313 B2 2/2009 Toshima et al.
7,507,325 B2 3/2009 Gueret et al.
7,513,989 B1 4/2009 Soled et al.
7,534,342 B2 5/2009 Bhan et al.
7,585,406 B2 9/2009 Khadzhiev et al.
7,588,681 B2 9/2009 Bhan et al.
7,651,604 B2 1/2010 Ancheyta Juarez et al.
7,651,605 B2 1/2010 Sahara et al.
7,695,610 B2 4/2010 Bolshakov et al.
7,713,905 B2 5/2010 Dufresne et al.
7,718,050 B2 5/2010 Gueret et al.
7,754,162 B2 7/2010 Dassori
7,901,569 B2 3/2011 Farshid et al.
7,938,955 B2 5/2011 Araki et al.
7,943,035 B2 5/2011 Chornet et al.
8,012,343 B2 9/2011 Plantenga et al.
8,021,538 B2 9/2011 Klein
8,114,806 B2 2/2012 Bhan et al.
8,163,166 B2 4/2012 Wellington et al.
8,173,570 B2 5/2012 Maesen et al.
8,241,489 B2 8/2012 Bhan et al.
8,268,164 B2 9/2012 Wellington et al.
8,318,000 B2 11/2012 Bhan et al.
8,318,628 B2 11/2012 Brun et al.
8,343,887 B2 1/2013 Maesen et al.
8,372,268 B2 2/2013 Ginestra et al.
8,394,254 B2 3/2013 Wellington et al.
8,394,262 B2 3/2013 Guichard et al.
8,475,651 B2 7/2013 Milam et al.
8,506,794 B2 8/2013 Bhan et al.
8,546,626 B2 10/2013 Daudin et al.
8,563,456 B2 10/2013 Dillon et al.
8,608,938 B2 12/2013 Wellington et al.
8,608,946 B2 12/2013 Bhan et al.
8,613,851 B2 12/2013 Wellington et al.
8,652,817 B2 2/2014 Wood et al.
8,663,453 B2 3/2014 Wellington et al.
8,679,319 B2 3/2014 Milam et al.
8,679,322 B2 3/2014 Marzin et al.
8,702,970 B2 4/2014 Maesen et al.
8,716,164 B2 5/2014 Dillon et al.
8,721,871 B1 5/2014 Dindi
8,722,558 B2 5/2014 Konno et al.
8,722,563 B2 5/2014 Soled et al.
8,722,564 B2 5/2014 Soled et al.
8,741,129 B2 6/2014 Brown et al.
8,747,659 B2 6/2014 Kiss et al.
8,764,972 B2 7/2014 Bhan et al.
8,784,646 B2 7/2014 Sanchez et al.
8,795,514 B2 8/2014 Kimura et al.
8,821,714 B2 9/2014 Chaumonnot et al.
8,877,040 B2 11/2014 Hoehn et al.
8,894,838 B2 11/2014 Dindi et al.
8,926,826 B2 1/2015 Dindi et al.
8,946,110 B2 2/2015 Toledo Antonio et al.
8,962,514 B2 2/2015 Seki et al.
8,987,537 B1 3/2015 Droubi et al.
8,999,011 B2 4/2015 Stern et al.
9,057,035 B1 6/2015 Kraus et al.
9,102,884 B2 8/2015 Xu et al.
9,109,176 B2 8/2015 Stern et al.
9,127,215 B2 9/2015 Choi et al.
9,127,218 B2 9/2015 Banerjee et al.
9,139,782 B2 9/2015 Dindi et al.
9,206,363 B2 12/2015 Woo et al.
9,212,323 B2 12/2015 Dindi et al.
9,216,407 B2 12/2015 Duma et al.
9,260,671 B2 2/2016 Shafi et al.
9,278,339 B2 3/2016 Bellussi et al.
9,340,733 B2 5/2016 Marchand et al.
9,359,561 B2 6/2016 Bazer-Bachi et al.
9,365,781 B2 6/2016 Dindi
9,365,782 B2 6/2016 Dindi et al.
9,387,466 B2 7/2016 Rana et al.
9,434,893 B2 9/2016 Dufresne
9,458,396 B2 10/2016 Weiss
9,487,718 B2 11/2016 Kraus et al.
9,499,758 B2 11/2016 Droubi et al.
9,512,319 B2 12/2016 Chatron-Michuad et al.
9,540,573 B2 1/2017 Bhan
9,546,327 B2 1/2017 Krasu et al.
9,605,215 B2 3/2017 Lott et al.
9,624,448 B2 4/2017 Joo et al.
9,650,312 B2 5/2017 Baldassari et al.
9,650,580 B2 5/2017 Merdrignac et al.
9,657,236 B2 5/2017 Yang et al.
9,675,968 B2 6/2017 Alonso Nunez et al.
9,737,883 B2 8/2017 Yamane et al.
9,896,630 B2 2/2018 Weiss et al.
9,908,105 B2 3/2018 Duma et al.

(56) **References Cited**

U.S. PATENT DOCUMENTS

9,908,107	B2	3/2018	Osaki et al.
9,919,293	B1	3/2018	Rana et al.
10,138,438	B2	11/2018	Van Houten
10,144,882	B2	12/2018	Dindi et al.
10,150,930	B2	12/2018	Van Houten
10,260,009	B2	4/2019	Ackerson et al.
10,443,006	B1	10/2019	Fruchey et al.
10,563,133	B2	2/2020	Moore et al.
10,597,594	B1	3/2020	Fruchey et al.
10,717,938	B2	7/2020	Ackerson et al.
10,800,982	B2	10/2020	Peer et al.
10,836,966	B2 *	11/2020	Moore C10G 45/04
10,899,983	B1	1/2021	Kar et al.
10,954,456	B2	3/2021	Moore et al.
10,961,468	B2	3/2021	Moore et al.
10,995,290	B2	5/2021	Anderson et al.
11,021,662	B2 *	6/2021	Moore C10L 1/04
11,118,120	B2	9/2021	Brown et al.
11,118,122	B2	9/2021	Ramaseshan et al.
11,124,714	B2	9/2021	Eller et al.
11,149,217	B2	10/2021	Marques et al.
11,168,264	B2	11/2021	Brahem et al.
11,203,724	B2	12/2021	Pereira Almao et al.
11,236,281	B2	2/2022	Rogel et al.
11,261,387	B2	3/2022	Wei et al.
11,345,865	B2	5/2022	Zhang et al.
11,384,301	B2	7/2022	Eller et al.
11,396,633	B2	7/2022	Kar et al.
11,421,166	B2	8/2022	Weiss et al.
11,459,514	B2	10/2022	Pupat
11,466,222	B2	10/2022	Markkanen
2001/0001036	A1	5/2001	Espeillac et al.
2001/0013484	A1	8/2001	Zeuthen et al.
2002/0037806	A1	3/2002	Dufresne et al.
2002/0045540	A1	4/2002	Bartholdy
2002/0056664	A1	5/2002	Chabot
2002/0070147	A1	6/2002	Sonnemans et al.
2002/0117426	A1	8/2002	Holder
2002/0144932	A1	10/2002	Gong et al.
2002/0148757	A1	10/2002	Huff et al.
2002/0157990	A1	10/2002	Feimer et al.
2002/0195375	A1	12/2002	Chapus et al.
2003/0042172	A1	3/2003	Sharivker et al.
2003/0125198	A1	7/2003	Ginestra et al.
2003/0131526	A1	7/2003	Kresnyak et al.
2003/0146133	A1	8/2003	Nagamatsu et al.
2003/0217951	A1	11/2003	Marchal-George et al.
2004/0007501	A1	1/2004	Sughrue et al.
2004/0020829	A1	2/2004	Magna et al.
2004/0040890	A1	3/2004	Morton et al.
2004/0055934	A1	3/2004	Tromeur et al.
2004/0134837	A1	7/2004	Dassori
2004/0178117	A1	9/2004	Morton et al.
2004/0186014	A1	9/2004	Tsukada et al.
2004/0209771	A1	10/2004	Abe et al.
2004/0232041	A1	11/2004	Kiser et al.
2004/0256293	A1	12/2004	Abe et al.
2005/0020446	A1	1/2005	Choudhary et al.
2005/0101480	A1	5/2005	Ackerman et al.
2005/0109674	A1	5/2005	Klein
2005/0113250	A1	5/2005	Schleicher et al.
2005/0133405	A1	6/2005	Wellington et al.
2005/0133406	A1	6/2005	Wellington et al.
2005/0133411	A1	6/2005	Zeuthen et al.
2005/0133416	A1	6/2005	Bhan et al.
2005/0133417	A1	6/2005	Bhan et al.
2005/0135997	A1	6/2005	Wellington et al.
2005/0139512	A1	6/2005	Wellington et al.
2005/0139520	A1	6/2005	Bhan et al.
2005/0139522	A1	6/2005	Bhan et al.
2005/0145537	A1	7/2005	Wellington et al.
2005/0145538	A1	7/2005	Wellington et al.
2005/0145543	A1	7/2005	Bhan et al.
2005/0148487	A1	7/2005	Brownscombe et al.
2005/0150156	A1	7/2005	Karas et al.
2005/0150818	A1	7/2005	Bhan et al.
2005/0155906	A1	7/2005	Wellington et al.
2005/0167321	A1	8/2005	Wellington et al.
2005/0167327	A1	8/2005	Bhan et al.
2005/0167328	A1	8/2005	Bhan et al.
2005/0167329	A1	8/2005	Bhan et al.
2005/0167331	A1	8/2005	Bhan et al.
2005/0269245	A1	12/2005	Huve
2006/0052235	A1	3/2006	Bai et al.
2006/0060501	A1	3/2006	Gauthier et al.
2006/0060509	A1	3/2006	Miyauchi et al.
2006/0060510	A1	3/2006	Bhan
2006/0102522	A1	5/2006	Turaga et al.
2006/0115392	A1	6/2006	Dassori
2006/0175229	A1	8/2006	Montanari et al.
2006/0211900	A1	9/2006	Iki et al.
2006/0231456	A1	10/2006	Bhan
2006/0231465	A1	10/2006	Bhan
2006/0234876	A1	10/2006	Bhan
2006/0234877	A1	10/2006	Bhan
2006/0249429	A1	11/2006	Iki et al.
2006/0281638	A1	12/2006	Zaid et al.
2006/0289340	A1	12/2006	Brownscombe et al.
2007/0000808	A1	1/2007	Bhan et al.
2007/0000810	A1	1/2007	Bhan et al.
2007/0012595	A1	1/2007	Brownscombe et al.
2007/0072765	A1	3/2007	Soled et al.
2007/0084753	A1	4/2007	Iki et al.
2007/0105714	A1	5/2007	Turaga et al.
2007/0108098	A1	5/2007	Flint et al.
2007/0131584	A1	6/2007	Kalnes
2007/0138055	A1	6/2007	Farshid et al.
2007/0170096	A1	7/2007	Shan et al.
2007/0175797	A1	8/2007	Iki et al.
2007/0284285	A1	12/2007	Stepanik et al.
2008/0017551	A1	1/2008	Kiriyama et al.
2008/0047875	A1	2/2008	Karas et al.
2008/0073247	A1	3/2008	Bolshakov et al.
2008/0085225	A1	4/2008	Bhan et al.
2008/0135453	A1	6/2008	Bhan
2008/0149531	A1	6/2008	Roy-Auberger et al.
2008/0167180	A1	7/2008	Van Den Brink et al.
2008/0210595	A1	9/2008	Bolshakov et al.
2008/0223755	A1	9/2008	Roy-Auberger et al.
2008/0230440	A1	9/2008	Graham et al.
2008/0245700	A1	10/2008	Wellington et al.
2008/0245702	A1	10/2008	Wellington et al.
2008/0262115	A1	10/2008	Calis et al.
2008/0272027	A1	11/2008	Wellington et al.
2008/0272028	A1	11/2008	Wellington et al.
2008/0308459	A1	12/2008	Iki et al.
2009/0048097	A1	2/2009	Jones et al.
2009/0057194	A1	3/2009	Farshid et al.
2009/0057197	A1	3/2009	Bhan et al.
2009/0062590	A1	3/2009	Nadler et al.
2009/0114569	A1	5/2009	Saheni et al.
2009/0134064	A1	5/2009	Reynolds
2009/0139902	A1	6/2009	Kressmann et al.
2009/0166260	A1	7/2009	Magalie et al.
2009/0178951	A1	7/2009	Balthasar et al.
2009/0230022	A1	9/2009	Gorbaty et al.
2009/0234166	A1	9/2009	Gorbaty et al.
2009/0255850	A1	10/2009	Bhan et al.
2009/0255851	A1	10/2009	Bhan et al.
2009/0275788	A1	11/2009	Bedard et al.
2009/0283444	A1	11/2009	Bhan et al.
2009/0288987	A1	11/2009	Bhan et al.
2009/0308791	A1	12/2009	Bhan et al.
2009/0308812	A1	12/2009	Osaheni et al.
2009/0314686	A1	12/2009	Zimmerman
2010/0006475	A1	1/2010	Ginestra
2010/0018902	A1	1/2010	Brownscombe et al.
2010/0025291	A1	2/2010	Shafi et al.
2010/0044274	A1	2/2010	Brun et al.
2010/0055005	A1	3/2010	Bhan et al.
2010/0098602	A1	4/2010	Bhan et al.
2010/0155301	A1	6/2010	Guichard et al.
2010/0200463	A1	8/2010	Patron et al.
2010/0213103	A1	8/2010	Patron et al.

(56) **References Cited**

U.S. PATENT DOCUMENTS

2010/0243526 A1 9/2010 Ginestra et al.
2010/0243532 A1 9/2010 Myers et al.
2010/0264067 A1 10/2010 Osaheni et al.
2010/0294698 A1 11/2010 E Mello et al.
2010/0326890 A1 12/2010 Bhan
2011/0017637 A1 1/2011 Reynolds et al.
2011/0079542 A1 4/2011 Ellis et al.
2011/0083997 A1 4/2011 Silva et al.
2011/0094938 A1 4/2011 Morel
2011/0108461 A1 5/2011 Gabrielov et al.
2011/0127194 A1 6/2011 Zhang et al.
2011/0155558 A1 6/2011 Cardoso et al.
2011/0155644 A1 6/2011 Bhattacharyya et al.
2011/0174681 A1 7/2011 Milam et al.
2011/0178346 A1 7/2011 Milam et al.
2011/0186477 A1 8/2011 Milam et al.
2011/0186480 A1 8/2011 Milam et al.
2011/0203971 A1 8/2011 Kiss et al.
2011/0218097 A1 9/2011 Rayo Mayoral et al.
2011/0240517 A1 10/2011 Chornet et al.
2011/0277377 A1 11/2011 Novak et al.
2012/0018352 A1 1/2012 Seki et al.
2012/0103868 A1 5/2012 Dindi et al.
2012/0116145 A1 5/2012 Bhan et al.
2012/0145528 A1 6/2012 Myers et al.
2012/0175285 A1 7/2012 Bhan et al.
2012/0175286 A1 7/2012 Bhan et al.
2012/0181219 A1 7/2012 Seki et al.
2013/0037447 A1 2/2013 Zimmerman
2013/0081977 A1 4/2013 Woo et al.
2013/0105357 A1 5/2013 Bhan
2013/0105364 A1 5/2013 Bhan
2013/0126393 A1 5/2013 Ginestra et al.
2013/0171039 A1 7/2013 Graham et al.
2013/0186806 A1 7/2013 Diehl et al.
2013/0225400 A1 8/2013 Liang et al.
2013/0256190 A1 10/2013 Van Wees et al.
2013/0267409 A1 10/2013 Lee et al.
2013/0277273 A1 10/2013 Mazyar
2013/0288885 A1 10/2013 Domokos et al.
2013/0306517 A1 11/2013 Kester et al.
2014/0027351 A1 1/2014 Bazer-Bachi et al.
2014/0061094 A1 3/2014 Xu et al.
2014/0073821 A1 3/2014 Mitsui et al.
2014/0076783 A1 3/2014 Bhan
2014/0097125 A1 4/2014 Bazer-Bachi et al.
2014/0166540 A1 6/2014 Guichard et al.
2014/0174980 A1 6/2014 Brown et al.
2014/0174983 A1 6/2014 Klein et al.
2014/0183098 A1 7/2014 Cooper et al.
2014/0183099 A1 7/2014 Ginestra et al.
2014/0291203 A1 10/2014 Molinari et al.
2014/0305843 A1 10/2014 Kraus et al.
2014/0315712 A1 10/2014 Smegal
2014/0323779 A1 10/2014 Alphazan et al.
2014/0326642 A1 11/2014 Tanaka et al.
2014/0353210 A1 12/2014 Graham et al.
2015/0057205 A1 2/2015 Morishima et al.
2015/0108039 A1 4/2015 Bhan
2015/0111726 A1 4/2015 Bhan et al.
2015/0144531 A1 5/2015 Ginstra et al.
2015/0144532 A1 5/2015 He et al.
2015/0217261 A1 8/2015 Norling
2015/0224476 A1 8/2015 Plecha et al.
2015/0240174 A1 8/2015 Kraus et al.
2015/0275105 A1* 10/2015 Touzalin C10G 45/32
585/260
2015/0315480 A1 11/2015 Hanks et al.
2015/0321177 A1 11/2015 Rana et al.
2015/0337225 A1 11/2015 Droubi et al.
2015/0353848 A1 12/2015 Patron
2015/0353851 A1 12/2015 Buchanan
2016/0001272 A1 1/2016 Daudin
2016/0017240 A1 1/2016 Duma et al.
2016/0024396 A1 1/2016 Zink et al.
2016/0060549 A1 3/2016 Ancheyta Juarez et al.
2016/0074840 A1 3/2016 Duma et al.
2016/0075954 A1 3/2016 Monson et al.
2016/0122666 A1 5/2016 Weiss et al.
2016/0129428 A1 5/2016 Bhan
2016/0145503 A1 5/2016 Xu et al.
2016/0145508 A1 5/2016 Xu et al.
2016/0145509 A1 5/2016 Mukherjee et al.
2016/0152901 A1 6/2016 Dufresne
2016/0160139 A1* 6/2016 Robinson C10G 45/02
208/15
2016/0177205 A1 6/2016 Evans et al.
2016/0200990 A1 7/2016 Mori et al.
2016/0220985 A1 8/2016 Osaki et al.
2016/0220986 A1 8/2016 Osaki et al.
2016/0230102 A1 8/2016 Osaki et al.
2016/0243528 A1 8/2016 He et al.
2016/0250622 A1 9/2016 He et al.
2016/0256856 A1 9/2016 Kester et al.
2016/0264887 A1 9/2016 Davydov
2016/0304794 A1 10/2016 Majcher et al.
2016/0340597 A1 11/2016 Baldassari et al.
2016/0348012 A1 12/2016 Zhao et al.
2016/0348013 A1 12/2016 Ladkat et al.
2016/0362615 A1 12/2016 Ancheyta Juarez et al.
2017/0002273 A1 1/2017 Rubin-Pitel et al.
2017/0002279 A1 1/2017 Brown et al.
2017/0009163 A1 1/2017 Kraus et al.
2017/0022433 A1 1/2017 Brown et al.
2017/0029723 A1 2/2017 Bazer-Bachi et al.
2017/0037325 A1 2/2017 Ackerson et al.
2017/0044451 A1 2/2017 Kar et al.
2017/0058205 A1 3/2017 Ho et al.
2017/0058223 A1 3/2017 Proubi et al.
2017/0066979 A1 3/2017 Lei et al.
2017/0073592 A1 3/2017 Nonaka et al.
2017/0120224 A1 5/2017 Boualleg et al.
2017/0120228 A1 5/2017 Boualleg et al.
2017/0120229 A1 5/2017 Boualleg et al.
2017/0121612 A1 5/2017 Boualleg et al.
2017/0128912 A1 5/2017 Boualleg et al.
2017/0136446 A1 5/2017 Carati et al.
2017/0137725 A1 5/2017 Boualleg et al.
2017/0165639 A1 6/2017 Klein et al.
2017/0175012 A1 6/2017 Schleiffer et al.
2017/0183575 A1 6/2017 Rubin-Pitel et al.
2017/0183582 A1 6/2017 Hoehn et al.
2017/0192126 A1 7/2017 Koseoglu
2017/0260463 A1 9/2017 Schleiffer et al.
2017/0267937 A1 9/2017 Schleiffer et al.
2017/0306250 A1 10/2017 Ginestra
2017/0306252 A1 10/2017 Malek Abbaslou et al.
2017/0335206 A1 11/2017 Mukherjee et al.
2017/0349846 A1 12/2017 Ding et al.
2017/0355913 A1 12/2017 Mountainland et al.
2017/0355914 A1 12/2017 Weiss et al.
2017/0362514 A1 12/2017 Hanks et al.
2018/0016505 A1 1/2018 Matsushita
2018/0104676 A1 4/2018 Yamane et al.
2018/0134965 A1 5/2018 Brown et al.
2018/0134972 A1 5/2018 Brown et al.
2018/0134974 A1 5/2018 Weiss et al.
2018/0147567 A1 5/2018 Matsushita et al.
2018/0154340 A1 6/2018 Boualleg et al.
2018/0230388 A1 8/2018 Li et al.
2018/0230389 A1* 8/2018 Moore C10L 1/1608
2018/0251690 A1 9/2018 Mountainland et al.
2018/0291291 A1 10/2018 Brown et al.
2018/0371343 A1 12/2018 Rubin-Patel et al.
2019/0002772 A1 1/2019 Moore et al.
2019/0010405 A1 1/2019 Moore et al.
2019/0016972 A1* 1/2019 Moore C10L 1/12
2019/0078027 A1 3/2019 Deimund et al.
2019/0093026 A1 3/2019 Wohaibi et al.
2019/0136144 A1 5/2019 Wohaibi et al.
2019/0185772 A1 6/2019 Berkhous et al.
2019/0203130 A1 7/2019 Mukherjee
2019/0233732 A1 8/2019 Sun
2019/0338203 A1 11/2019 Umansky et al.

(56) **References Cited**

U.S. PATENT DOCUMENTS

2019/0338205	A1	11/2019	Ackerson et al.	
2019/0382668	A1	12/2019	Klussmann et al.	
2020/0095508	A1	3/2020	Moore et al.	
2020/0140765	A1*	5/2020	Moore	C10G 67/06
2020/0224108	A1	7/2020	Moore et al.	
2020/0231886	A1	7/2020	Kraus et al.	
2020/0248080	A1	8/2020	Peer et al.	
2020/0291317	A1	9/2020	Anderson et al.	
2020/0339894	A1	10/2020	Marques et al.	
2020/0385644	A1	12/2020	Rogel et al.	
2021/0024842	A1	1/2021	Fruchey et al.	
2021/0062096	A1	3/2021	Hodgkins et al.	
2021/0102130	A1	4/2021	Marques et al.	
2021/0155858	A1	5/2021	Koseoglu et al.	
2021/0238487	A1	8/2021	Moore et al.	
2021/0246391	A1	8/2021	Anderson et al.	
2021/0253960	A1	8/2021	Eller et al.	
2021/0253964	A1	8/2021	Eller et al.	
2021/0253965	A1	8/2021	Woodchick et al.	
2021/0284919	A1	9/2021	Moore et al.	
2021/0292661	A1	9/2021	Klussmann et al.	
2021/0363434	A1	11/2021	Zhang et al.	
2021/0363444	A1	11/2021	Kar et al.	
2022/0073830	A1	3/2022	Harada et al.	
2022/0235275	A1	7/2022	Iversen et al.	
2022/0333020	A1*	10/2022	Moore	B01J 8/008
2022/0381547	A1*	12/2022	Moore	C10G 65/04
2022/0411702	A1	12/2022	Sarjovaara et al.	

FOREIGN PATENT DOCUMENTS

CA	1238005	A	6/1988
CA	1248513		1/1989
EP	804288	A1	11/1997
EP	870817	A1	10/1998
EP	B76443	A1	11/1998
EP	952888	A1	11/1999
EP	1041133	A1	10/2000
EP	1050572	A2	11/2000
EP	1052015	A1	11/2000
EP	1299192	A1	4/2003
EP	1358302	A0	11/2003
EP	1352946	A4	12/2004
EP	1567262	A1	8/2005
EP	1709141	A2	10/2006
EP	1894625	A2	3/2008
EP	2130895	A1	12/2009
EP	2167616	A1	3/2010
EP	2510076	A2	10/2012
EP	2907867	A1	8/2015
EP	2947133	A1	11/2015
EP	2947135	A1	11/2015
EP	2978824	A0	2/2016
EP	2992070	A2	3/2016
EP	3041608	A1	7/2016
EP	3074485	A0	10/2016
EP	2990465	A1	3/2019
FR	2681871		4/1993
FR	3011004	A1	3/2015
FR	3013723	A1	5/2015
FR	3053356		1/2018
GB	1324167		7/1973
GB	1502915		3/1978
GB	1504586		3/1978
GB	1505886		3/1978
GB	2121252	B	6/1986
JP	4801344	B2	10/2011
JP	2015059220	A	3/2015
JP	20200122450		8/2020
RU	2700705		9/2019
WO	9113951	A1	9/1991
WO	9820969		5/1998
WO	9919061	A1	4/1999
WO	200145839	A1	6/2001
WO	0197971	A1	12/2001
WO	0209870	A2	2/2002
WO	02062926	A2	8/2002
WO	2004052534	A1	6/2004
WO	2004053028	A1	6/2004
WO	2005028596	A1	3/2005
WO	2005063933	A2	7/2005
WO	2009001314	A1	12/2008
WO	2011071705	A2	6/2011
WO	2014096703	A1	6/2014
WO	2014160603	A1	10/2014
WO	2014177424	A2	11/2014
WO	201534521	A1	3/2015
WO	2015078674	A1	6/2015
WO	2015097199	A1	7/2015
WO	2015122931	A1	8/2015
WO	2015147222	A1	10/2015
WO	2015147223	A1	10/2015
WO	2015178941	A1	11/2015
WO	2015179017	A2	11/2015
WO	2015189190	A1	12/2015
WO	2016089590	A1	6/2016
WO	2016146326	A1	9/2016
WO	2016195973	A1	12/2016
WO	201780387	A1	5/2017
WO	2017186484	A1	11/2017
WO	2018073018	A1	4/2018
WO	2018075015	A1	4/2018
WO	2018075016	A1	4/2018
WO	2018075017	A1	4/2018
WO	2018093535	A1	5/2018
WO	2018101244	A1	6/2018
WO	2018053323	A1	3/2019
WO	2019104243	A1	5/2019
WO	2019125674	A1	6/2019
WO	2019133880	A1	7/2019
WO	2019178701	A1	9/2019
WO	2020262078	A1	12/2020
WO	2021066265	A1	4/2021

OTHER PUBLICATIONS

Daniel Monzon, et al. "Petroleum refiners and shippers struggle over marine fuel", Author D Little opinion/ position paper, May 9, 2017, pp. 1-4, Arthur D. Little, https://www.adlittle.com/sites/default/files/viewpoints/adl_petroleum_refiners_and_shippers_struggle_over_marine_fuel.pdf.

Gulam Gaush Zeelani, et al. "Catalystic Oxidative Desulfurization of Liquid Fuel: A Review", International Research Journal of Engineering and Technology, May 2016 pp. 331-336, vol. 3; Issue 5, Fast Track Publications, Tamilnadu, India.

Kristin Rist Sorheim et al. "Characterization of Low Sulfur Fuel Oils (LSFO)—A new generation of marine fuel oils", Oct. 7, 2020, pp. 1-22 , Report No. OC2020 A-050, Project No. 302004929, Version 3.1, SINTEF Ocean AS, Trondheim, Norway.

S. Brynolf et al. "Compliance possibilities for the future ECA regulations through the use of abatement technologies or change of fuels", Transportation Research Part D May 2014, pp. 6-18, vol. 28, Elsevier Ltd. London United Kingdom.

Alun Lewis, "Composition, Properties and Classification of Heavy Fuel Oils" Third R&D Forum on High-density Oil Spill Response, Mar. 11, 2002, pp. 11-25, Intype, London United Kingdom.

Tong-Rong Ling et al. "Desulfurization of Vacuum Gasoil By MCM-41 Supported Molybdenum-Nickel" , Ind. Eng. Chem. Res. Jan. 14, 2009, vol. 48, pp. 1797-1803, American Chemical Society.

S.K. Maity et al. "Early Stage Deactivation of Heavy Crude Oil Hydroprocessing Catalysts", Fuel, vol. 100 (Nov. 23, 2011), pp. 17-23, Elsevier Ltd. London United Kingdom.

Kevin Cullinane et al. "Emission Control Areas and Their Impact on Maritime Transport", Transportation Research Part D May 2014, pp. 1-5, vol. 28, Elsevier Ltd. London United Kingdom.

Selma Brynolf, "Environmental Assessment of Present and Future Marine Fuels", Thesis for the Degree of Doctor of Philosophy, 2014, pp. 1-105, Department of Shipping and Marine Technology, Chalmers University of Technology SE-412 96 Gothenburg, Sweden.

(56) **References Cited**

OTHER PUBLICATIONS

Alexey Y. Kirgizov, et al. "Ex Situ Upgrading of Extra Heavy Oil: The Effect of Pore Shape of Co—Mo/-Al2O3 Catalysts", Catalysts Oct. 18, 2022, vol. 12, pp. 1271-1284, MDPI, Basel Switzerland https://doi.org/10.3390/catal12101271.
Sundaramurthy Vedachalam, et al., "Hydrotreating and oxidative desulfurization of heavy fuel oil into low sulfur marine fuel over dual function NiMo/γ-Al2O3 catalyst" Catalysis Today. vol. 407 (2023) pp. 165-171, available on-line Jan. 19, 2022, Elsevier Ltd. London United Kingdom.
Susana Trasobares, et al., "Kinetics of Conradson Carbon Residue Conversion in the Catalytic Hydroprocessing of a Maya Residue", Ind. Eng. Chem. Res. Jan. 5, 1998, vol. 37, pp. 11-17, American Chemical Society.
Jiang Zongxuan, et al., "Oxidative Desulfurization of Fuel Oils", Chin. J. Catal., 2011, vol. 32: pp. 707-715. Elsevier Ltd. London United Kingdom.
Sara Houda, et al., "Oxidative Desulfurization of Heavy Oils with High Sulfur Content: A Review", Catalysts Aug. 23, 2018, vol. 8, pp. 344-370. MDPI, Basel Switzerland , www.mdpi.com/journal/catalysts.
Fawzi M. Elfghi & N.A.S.AMIN, "Parametric Study of Hydrodesulfurization and Hydrodearomatization of Gasoil in Hydrotreating Process of Over CoMo—S Catalyst Using a Pilot Plant Integral Reactor", Jurnal Teknologi, Dec. 2011, vol. 56, pp. 53-73. Penerbit UTM Press, Universiti Teknologi Malaysia.
Jose Luis Garcia-Gutierrez, et al., "R & D in Oxidative Desulfurization of Fuels Technologies: From Chemistry to Patents", Recent Patents on Chemical Engineering, Dec. 2012, vol. 5, pp. 174-196.
Antoine Halff, et al., "The likely implications of the new IMO standards on the shipping industry", Energy Policy, Nov. 2018, vol. 126, pp. 277-286, Elsevier Ltd. London United Kingdom.
S. Houda, et al., "Ultrasound assisted oxidative desulfurization of marine fuels on MoO3/Al2O3 catalyst" Catalysis Today, Oct. 17, 2020, vol. 377, pp. 221-228, Elsevier Ltd. London United Kingdom.
Mohammad Farhat Ali, et al. A review of methods for the demetallization of residual fuel oils, Fuel Processing Technology, Mar. 8, 2006, pp. 573-584, vol. 87, Elsevier B.V.
Mohan S. Rana, et al., A review of recent advances on process technologies for upgrading of heavy oils and residua, Fuel, Available online Sep. 7, 2006, pp. 1216-1231, vol. 86, Elsevier B.V.
Carolina Leyva, et al., Activity and surface properties of NiMo/SiO2—Al2O3 catalysts for hydroprocessing of heavy oils, Applied Catalysis A: General, Available online Feb. 28, 2012, pp. 1-12, vol. 425-426, Elsevier B.V.
Oliver C. Mullins et al., Asphaltenes Explained for the Nonchemist, Petrophysics, Jun. 2015, pp. 266-275, vol. 56, No. 3.
M. Ghasemi, et al., Phase Behavior and Viscosity Modeling of Athabasca Bitumen and Light Solvent Mixtures, SPE165416, Jun. 2013, pp. 1-26, Society of Petroleum Engineers.
Marten Ternan, Catalytic Hydrogenation and Asphaltene Conversion of Athabasca Bitumen, The Canadian Journal of Chemical Engineering, Oct. 1983, pp. 689-696, vol. 61.
Jeremi'as Marti'nez, et al., Comparison of correlations to predict hydrotreating product properties during hydrotreating of heavy oils, Catalysis Today, Available online Dec. 5, 2009, pp. 300-307, vol. 150, Elsevier B.V.
Luis C. Castnteda, et al., Current situation of emerging technologies for upgrading of heavy oils, Catalysis Today, Available online Jul. 4, 2013, pp. 248-273, vol. 220-222, Elsevier B.V.
C. Ferreira, et al., Hydrodesulfurization and hydrodemetallization of different origin vacuum residues: Characterization and reactivity, Fuel, Available online Apr. 14, 2012, pp. 1-11, http://dx.doi.org/10.1016/j.fuel.2012.03.054, Elsevier B.V.
C.V. Philip, et al. GPC Characterization for Assessing Compatibility Problems with Heavy Fuel Oils, Fuel Processing Technology 1984: pp. 189-201., Elsevier B.V.
Muhammad A. Altajam & Marten Ternan, Hydrocracking of Athabasca bitumen using Co—Mo catalysts supported on wide pore carbon extrudates, Fuel, Aug. 1989, pp. 955-960, Butterworth & Co. Publishers Ltd.
J.W. Holmes & J.A. Bullin, Fuel Oil Compatibility Probed, Hydrocarbon Processing, Sep. 1983: pp. 101-103.
Charles J. Glover, & Jerry A. Bullin, Identification of Heavy Residual Oils by GC and GCMS, Journal of Environmental Science and Health A24(1), 1989: pp. 57-75.
H. Puron, et al., Kinetic analysis of vacuum residue hydrocracking in early reaction stages, Fuel, Available online Sep. 27, 2013, pp. 408-414, vol. 117, Elsevier B.V.
Yanet Villasna, et al. Upgrading and Hydrotreating of Heavy Oils and Residua, Energy Science and Technology, vol. 3, Oil and Natural Gas, 2015, pp. 304-328, Stadium Press LLC, Houston TX USA.
International Search Report issued in corresponding International Application No. PCT/US2018/017855 dated Apr. 27, 2018 (3 pages).
International Search Report issued in corresponding International Application No. PCT/US2018/017863 dated Apr. 27, 2018 (3 pages).
Tesoro Refining & Marketing Co. Material Safety Data Sheet—Fuel Oil, pp. 1-10, Jul. 26, 2012, San Antonio, Texas, US.
Tesoro Refining & Marketing Co. Material Safety Data Sheet—Marine Gas Oil, pp. 1-11, Nov. 17, 2012, San Antonio, Texas, US.
Tesoro Refining & Marketing Co. Material Safety Data Sheet—Resid pp. 1-10, Apr. 6, 2015, San Antonio, Texas, US.
Coutrymark Refining and Logistics, LLC, Material Safety Data Sheet—No. 6 Fuel Oil, Dec. 2012, pp. 1-4, Mt. Vernon, Indiana US.
Valero Marekting & Supply Company, Material Safety Data Sheet—Residual Fuel Oil, Dec. 4, 2010, pp. 1-14, San Antonio, Texas US.
Oceanbat SA. Material Safety Data Sheet—Marine Fuel Oil, Jul. 2013, pp. 1-7, Guayaquil Ecuador.
Exxonmobilcorporation, Material Safety Data Sheet—Marine Fuel Oil, pp. 1-12, Sep. 18, 2013, Fairfax Virginia US.
Shell Trading (US) Company, Material Safety Data Sheet—Ultra Low Sulfur Fuel Oil, pp. 1-21, Jun. 19, 2018, Houston, Texas US.
Suncor Energy Inc., Material Safety Data Sheet—Heating Fuel Oil Type 6 / Residual Marine Fuel, pp. 1-11, Jun. 7, 2018, Calgary Alberta Canada.
Marathon Petroleum Company LP, Material Safety Data Sheet—Marathon No. 6 Fuel Oil, Dec. 7, 2010, pp. 1-14., Findlay, Ohio US.
BP Australia Pty Ltd., Material Safety Data Sheet—BP380 Marine Fuel, Oct. 27, 2011. pages 1-6, Docklands, Victoria Australia.
U.S. Oil & Refining Co., Material Safety Data Sheet—Residual Fuel Oil, Dec. 18, 2008, pp. 1-11. Tacoma, Washington US.
American Bureau of Shipping, Publication 31 Notes On Heavy Fuel Oil, 1984, pp. 1-68, Houston Texas US.
American Bureau of Shipping, Notes Use of Low Sulphur Marine Fuel for Main and Auxiliary Diesel Engines, Jan. 1, 2010, pp. 1-68, Houston Texas US (https://www.eagle.org/eagleExternalPortalWEB/ShowProperty/BEA%20Repository/pdfs/Regulatory/Docs/LowSulphurNote_Engine).
Shuyi Zhang, Dong Liu, Wenan Deng, Guohe Que, A Review of Slurry-Phase Hydrocracking Heavy Oil Technology, Energy & Fuels, vol. 21, No. 6, Nov. 2007, pp. 3057-3062, American Chemical Society, Washington DC US.
Peiman Pourmoghaddam, Susan Davari, Zahra Delavar Moghaddam, A Technical And Economic Assessment of Fuel Oil Hydrotreating Technology For Steam Power Plant SO2 and NOx Emissions Control, Advances In Environmental Technology vol. 2, Issue 1, Accepted Oct. 3, 2016, pp. 45-54, Iranian Research Organization For Science and Technology, Tehran Islamic Republic of Iran.
Dawoud Bahzad, Jamal Al-Fadhli, Ayyad Al-Dhafeeri, Ali Abdal, Assessment of Selected Apparent Kinetic Parameters of the HDM and HDS reactions of Two Kuwaiti Residual Oils, Using Two Types of Commercial ARDS Catalysts, Energy & Fuels, vol. 24, Jan. 8, 2010, pp. 1495-1501, American Chemical Society, Washington DC US.
A. Marafi, A. Hauser, a Stanislaus, Atmospheric Residual Desulfurization Process for Residual Oil Upgrading: An Investigation of the Effect of Catalyst Type and Operation Severity on Product Oil Quality, Energy & Fuels, vol. 20, Apr. 4, 2006, pp. 1145-1149, American Chemical Society, Washington DC US.

(56) **References Cited**

OTHER PUBLICATIONS

M.M. Boduszynki, C.E. Rechsteiner, A.S.G. Shafzadeh, R.M.K. Carlson, Composition and Properties of Heavy Crudes, No. 1998. 202 UNITAR Centre for Heavy Crude and Tar Sands, 1998, pp. 1-12, Canada.
Gard AS, Bunkers and Bunkering—A selection of articles previously published by Gard AS, Jan. 2014, pp. 1-53, Arendal Norway.
Monique B. Vermeire Everything You Need to Know About Marine Fuels, Jun. 2012, pp. 1-32, Ghent Belgium.
T.M. Saleh, H. Ismail, J.E.Corbett, R.S. Bali, Commercial Experience in the Operation of Atmospheric Residue Desulfurization Unit in Kuwait national Petroleum Company At Mina Al-Ahmadi Refinery, Catalyst in Petroleum Refining, 1989, pp. 175-189, Elsevier Science Publishers B.V., Amsterdam The Netherlands.
Victor S. Semeykina, Ekaterina V. Parkhomchuk, Alexander V. Polukhin, Pavel D. Parunin, Anton I. Lysikov, Artem B. Ayupov, Svetlana V. Cherepanova, Vladislav V. Kanazhevskiy, Vasil V. Kaichev, Tatyana S. Glazneva, Valentina V. Zvereva, CoMoNi Catalyst Texture and Surface Properties in Heavy Oil Processing. Part I: Hierarchical Macro / Mesoporous Alumina Support, Industrial & Engineering Chemistry Research, vol. 55, Feb. 29, 2016, pp. 3535-3545 American Chemical Society, Washington DC US.
Victor S. Semeykina, Ekaterina V. Parkhomchuk, Alexander V. Polukhin, Pavel D. Parunin, Anton I. Lysikov, Artem B. Ayupov, Svetlana V. Cherepanova, Vladislav V. Kanazhevskiy, Vasil V. Kaichev, Tatyana S. Glazneva, Valentina V. Zvereva, CoMoNi Catalyst Texture and Surface Properties in Heavy Oil Processing. Part II: Macroporous Sepiolite-Like Mineral, Industrial & Engineering Chemistry Research, vol. 55,Aug. 1, 2016, pp. 9129-9139 American Chemical Society, Washington DC US.
Andre Hauser, Abdulazim Marafi, Adel Almutairi, Anthony Stanislaus, Comparative Study of Hydrodemetallization (HDM) Catalyst Aging by Boscan Feed and Kuwait Atmospheric Residue, Energy & Fuels, vol. 22 Aug. 27, 2008, pp. 2952-2932, American Chemical Society, Washington DC US.
Criterion Catalysts & Technologies LP, Residue Upgrading Product Information Sheet, pp. 1 & 2, Aug. 2008, Houston Texas US.
John-Laurent Tronche, Jelena Grigorjeva, Annie Siebert (editor), How Are Refiners Preparing For 2015 Marine Fuel Spec Changes?, pp. 1-2, Jun. 6, 2014, S&P Global Platts. Houston Texas US.
DNV GL Maritime, Hong Kong Requires Ocean-Going Vessels to Comply with 0.50% M/M Sulphur Limit While at Berth, Statutory Update No. 1, 2015 / March, p. 1, DNV GL Maritime, Hamburg Germany.
Mike Stockle, Tina Knight, Impact of Low-Sulphur Bunkers on Refineries, Catalysis 2009, p. 1-7, www.digitalrefining.com/article/1000090, article based on presentation from the Nov. 2008 ERC Annual Meeting, Vienna Austria.
Ekaterina V. Parkhomchuk, Anton I. Lysikov, Alexey G. Okunev, Pavel D. Parunin, Victoria S. Semeikina, Artem B. Ayupov, Valentina A. Trunova, Valentin N. Parmon, Meso / Macroporous CoMo Alumina Pellets for Hydrotreating of Heavy Oil, Industrial & Engineering Chemistry Research, vol. 15 Nov. 13, 2013, pp. 17117-17125 American Chemical Society, Washington DC US.
Cristian J. Calderon Jorge Ancheyta, Modeling of Slurry-Phase Reactors for Hydrocracking of Heavy Oils, Energy & Fuels, vol. 30 Jan. 28, 2016, pp. 2525-2543, American Chemical Society, Washington DC US.
DNV GL Maritime, Notice for Low Sulphur "Hybrid" Fuel Operation, Technical Update No. 3, 2015 / March, p. 1&2, DNV GL Maritime, Hamburg Germany.
Abdul Waheed Bhutto, Rashid Abro, Shurong Goa, Tauqeer Abbas, Xiaochun Chen, Guangren Yu, Oxidative Desulfurization of Fuel Oils Using Ionic Liquids: A Review, Journal of the Taiwan Institute of Chemical Engineers, vol. 62, Feb. 28, 2016, pp. 84-97, Elsevier B.V. Amsterdam The Netherlands.
V. Babich, J.A. Moulijn, Science and Technology of Novel Processes for Deep Desulfurization of Oil Refinery Streams: A Review, Fuel, vol. 82 , Nov. 14, 2002, pp. 607-631 Elsevier B.V. Amsterdam The Netherlands Published first on the web via fuelfirst.com—http://www.fuelfirst.com.
A. Hauser, A. Marafi, A. Stanislaus, A. Al-Adwani, Relation Between Feed Quality and Coke Formation in a Three Stage Atmospheric Residue Desulfurization (ARDS) Process, Energy & Fuels, vol. 19 Feb. 8, 2005, pp. 544-553, American Chemical Society, Washington DC US.
A Marafi, H. Al-Bazzaz, M. Al-Marri, F. Maruyama, M. Absi-Halbi, A. Stanislaus, Residual-Oil Hydrotreating Kinetics for graded Catalyst Systems: Effect of Original and Treated Feedstocks, Energy Fuels, vol. 17 (5), Jul. 2, 2003 pp. 1191-1197 American Chemical Society, Washington DC US.
Imaza Al-Bazza, Jia-Lin Kang, Dduha Chehadeh, Dawoud Bahzad, David Shan-Hill Wong, Shi-Shang Jang, Robust Predictions of Catalyst Deactivation of Atmospheric Residual Desulfurization, Energy Fuels, vol. 29, Oct. 21, 2015 pp. 7089-7100 American Chemical Society, Washington DC US.
A.G. Okunev, E.V. Parkhomchuk, A.I. Lysikov, P.D. Parunin, V.S. Semeykina, V.N. Parmon, Catalytic Hydroprocessing of Heavy Oil Feedstocks, Russian Chemical Reviews, vol. 84, Sep. 2015, pp. 981-999, Russian Academy of Sciences and Turpion Ltd. Moscow, Russia.
Ernest Czermanski, Slawomir Drozdziecki, Maciej Matczak, Eugen Spangenberg, Bogusz Wisnicki, Suplphur Regulation - Technology Solutions and Economic Consequences, Institute of Maritime transport and Seaborne Trade, University of Gdansk, 2014, pp. 1-76, University of Gdansk, Gdansk Poland.
Charles Olsen, Brian Watkins, Greg Rosinski, The Challenges of Processing FCC LCO, Catalagram 110 Special Edition, Fall 2011, pp. 6-8, W.R. Grace & Co. Advanced Refining Technologies, Columbus Maryland, US.
Yanzi Jia, Qinghe Yang, Shuling Sun, Hong Nie, Dadong Li, The Influence of Metal Deposits on Residue Hydrodemetallization Catalyst in the Absence and Presence of Coke, Energy Fuels, vol. 30 Feb. 22, 2016 pp. 2544-2554 American Chemical Society, Washington DC US.
James G. Speight, Upgrading Heavy Oils and Residua: The Nature of the Problem, Catalysis on the Energy Scence, 1984, pp. 515-527, Elsevier Science Publishers B.V. Amsterdam, The Netherlands.
Blessing Umana, Nan Zhang, Robin Smith, Development of Vacuum Residue Hydrodesuphurization—Hydrocracking Models and Their Integration with Refinery Hydrogen Networks, Industrial & Engineering Chemistry Research, vol. 55, Jan. 27, 2016, pp. 2391-2406, American Chemical Society, Washington DC US.
Mike Stockle, Tina Knight, Impact of Low Sulphur Bunkers on Refineries, Catalysis, 2009, pp. 1-7, downloaded from website: www.digitalrefining.com/article/1000090.
James G. Speight, The Desulfurization of Heavy Oils and Residua, 2nd Ed. 1999, Chapter 5, pp. 168-205, Marcel Dekker Inc. New York NY US.
James G. Speight, The Desulfurization of Heavy Oils and Residua, 2nd Ed. 1999, Chapter 6, pp. 206-253, Marcel Dekker Inc. New York NY US.
James G. Speight, The Desulfurization of Heavy Oils and Residua, 2nd Ed. 1999, Chapter 8, pp. 302-334, Marcel Dekker Inc. New York NY US.
James G. Speight, The Desulfurization of Heavy Oils and Residua, 2nd Ed. 1999, Chapter 9, pp. 335-385, Marcel Dekker Inc. New York NY US.

* cited by examiner

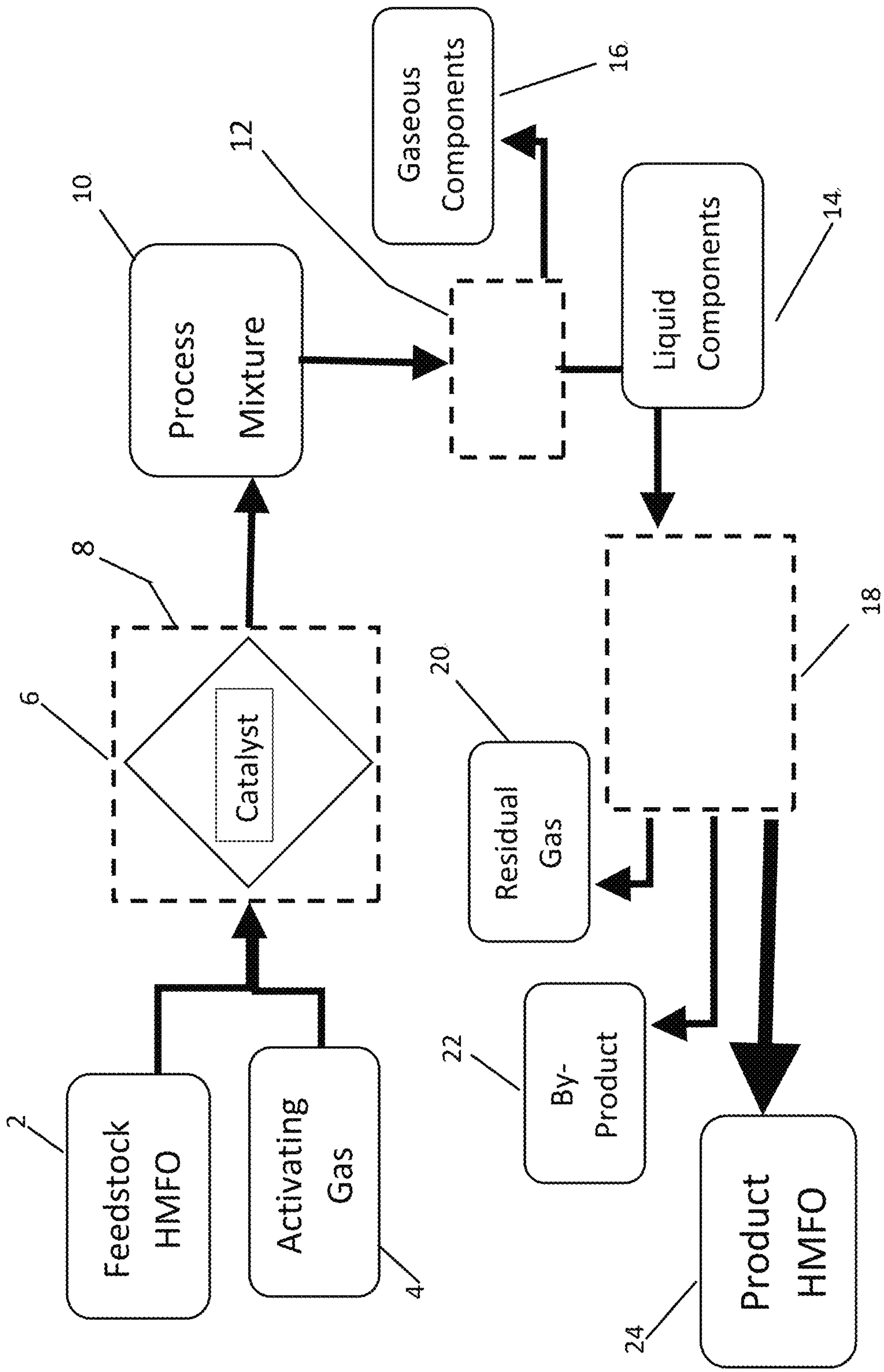
Feedstock HMFO
Activating Gas
Catalyst
Process Mixture
Gaseous Components
Liquid Components
Residual Gas
By-Product
Product HMFO
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FIG. 1

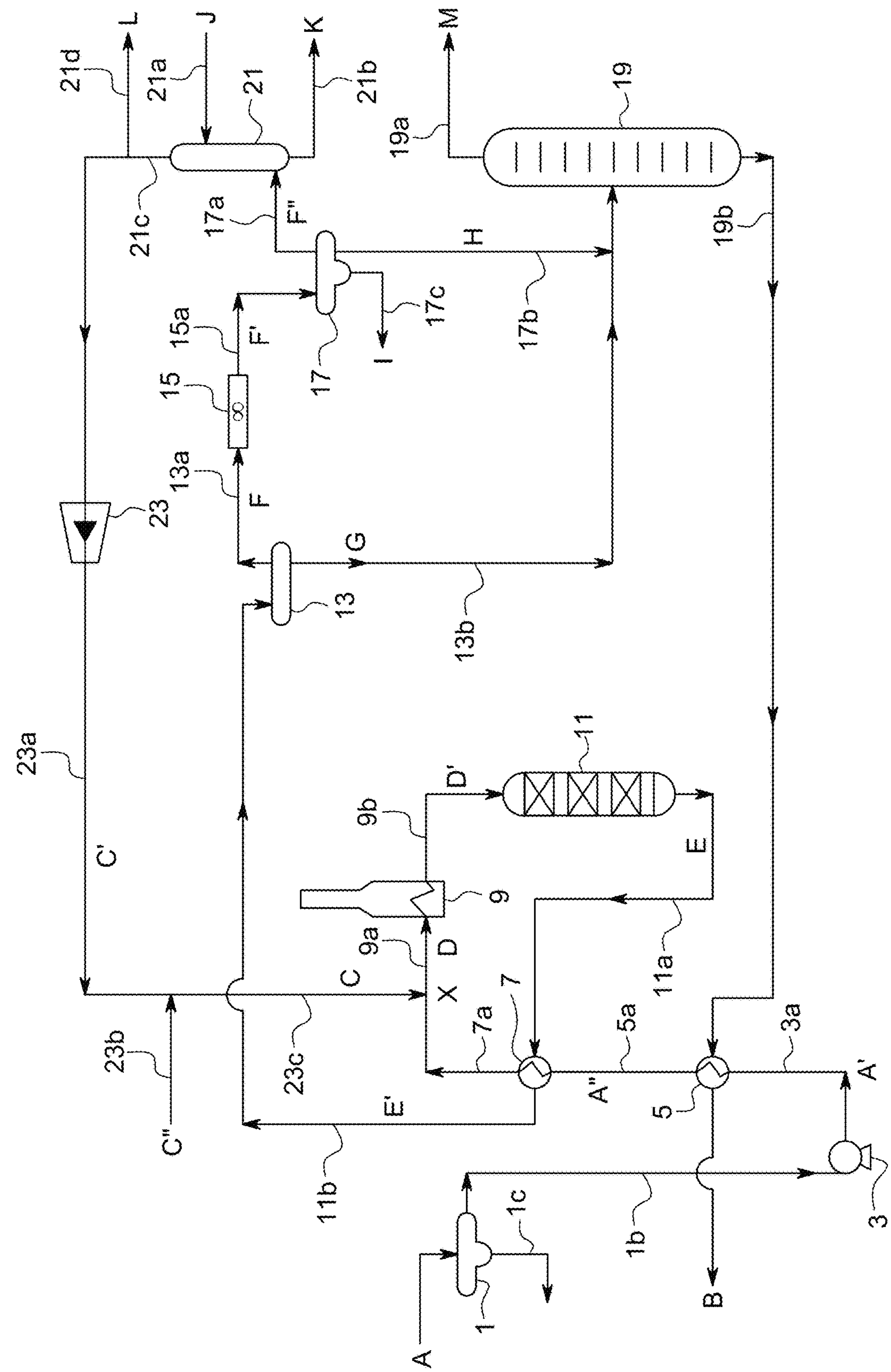

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FIG. 2

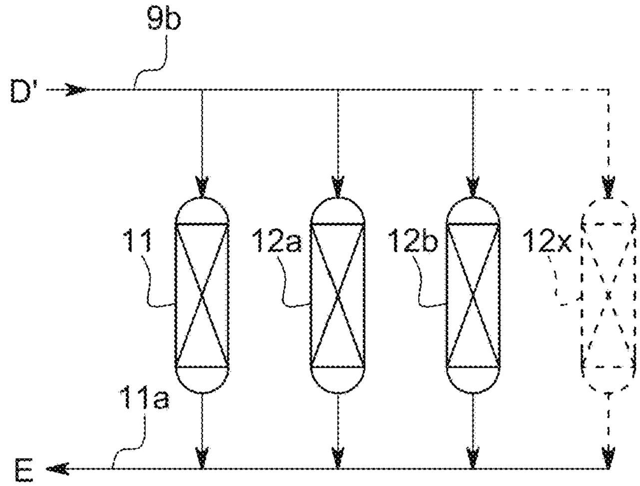


FIG. 3

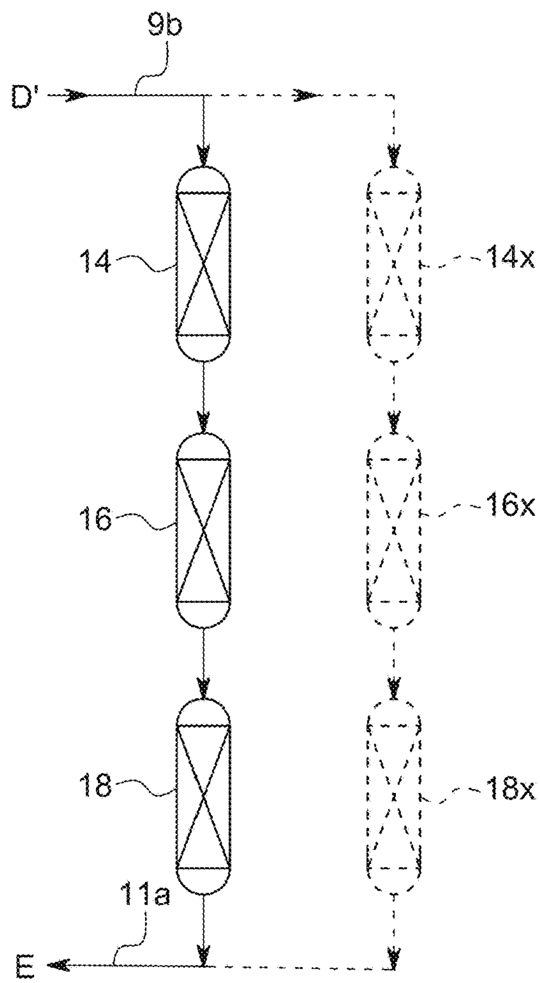


FIG. 4

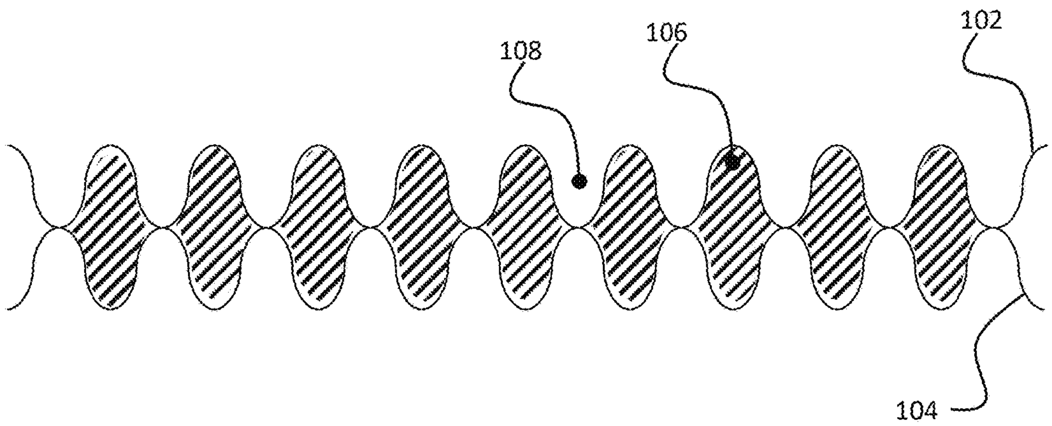


FIG. 5

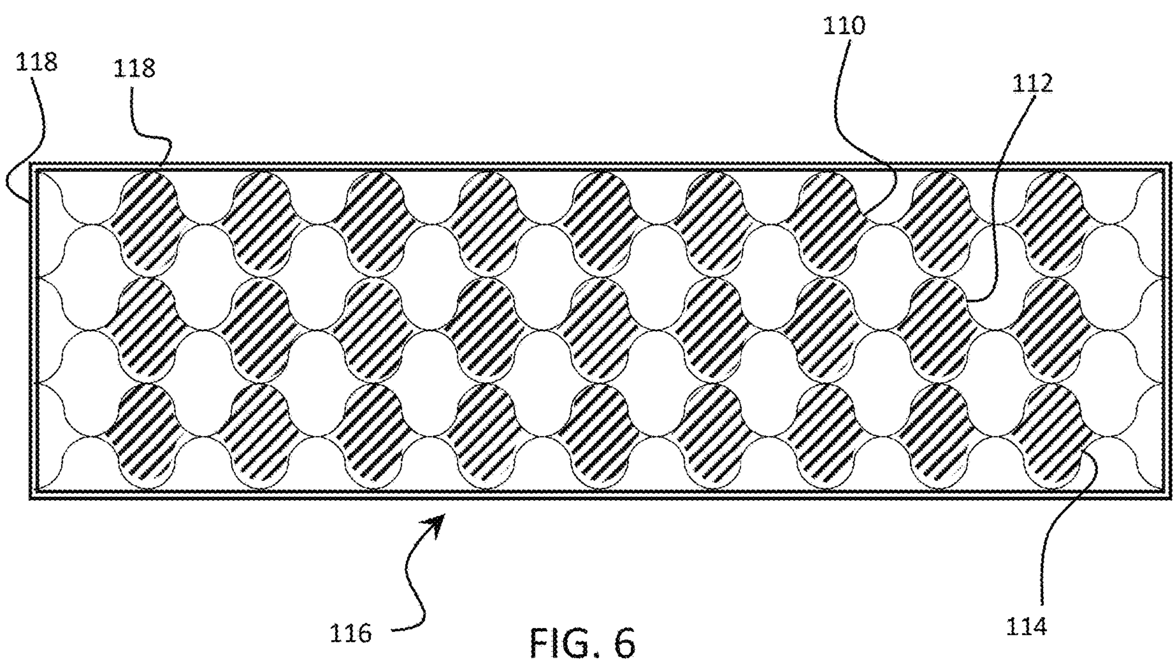


FIG. 6

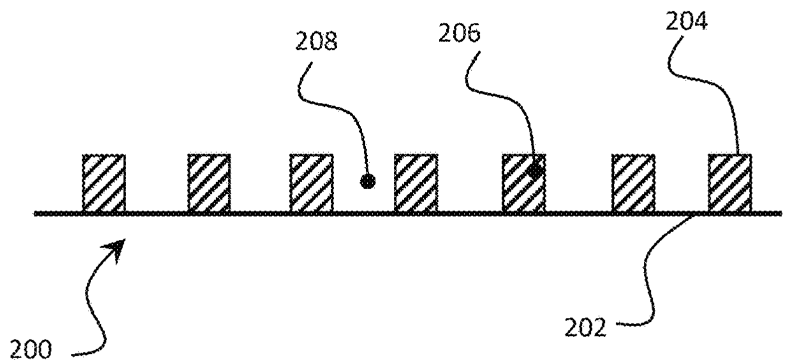


FIG. 7

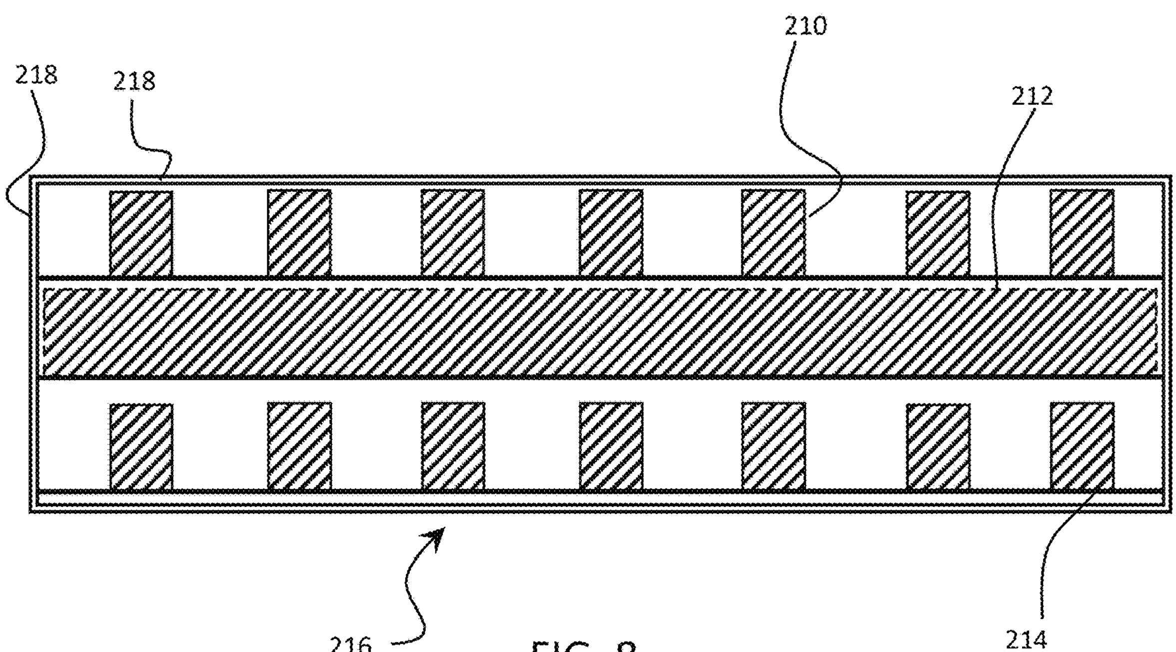
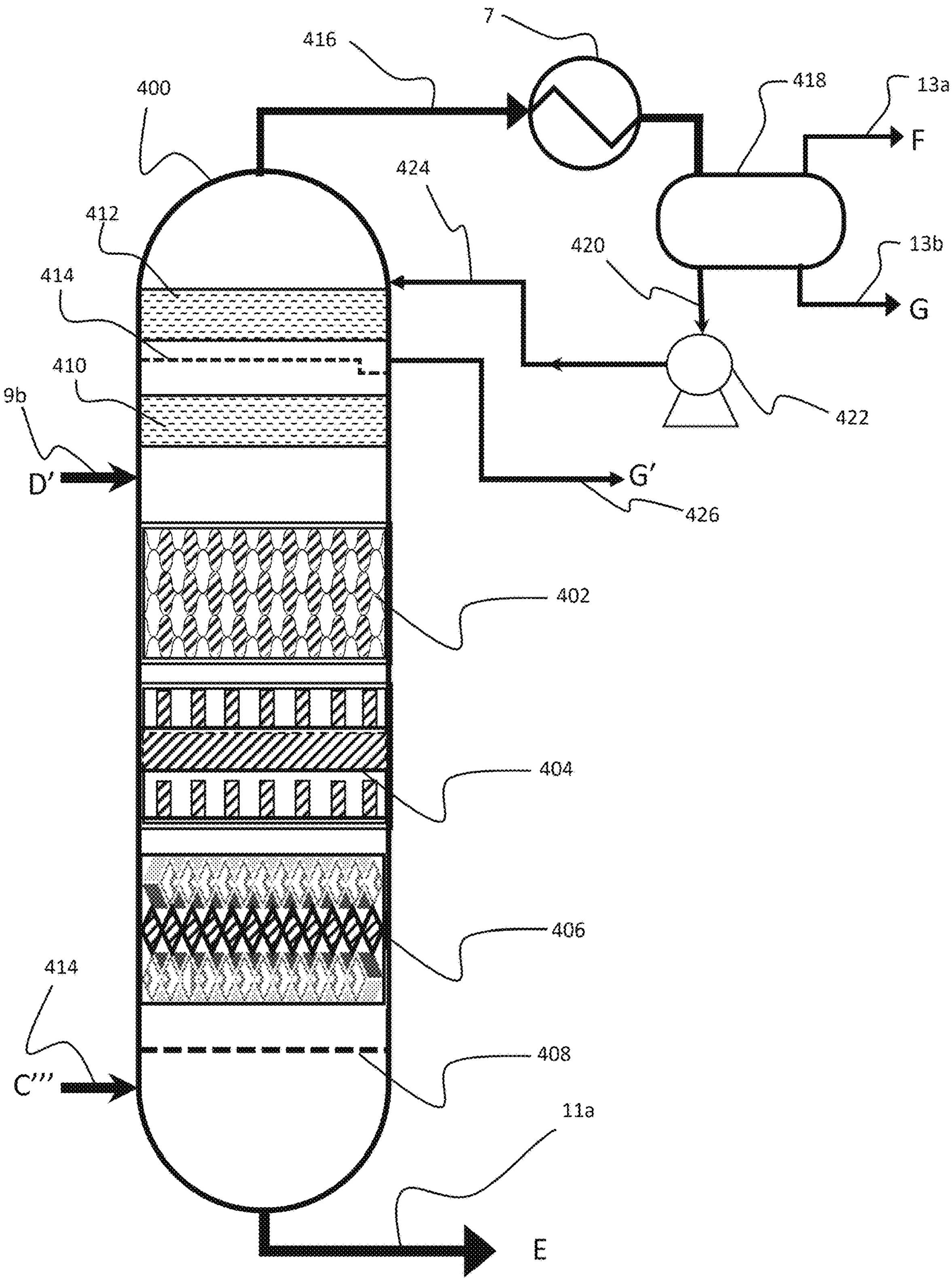


FIG. 8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11a
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FIG. 9

MULTI-STAGE PROCESS AND DEVICE UTILIZING STRUCTURED CATALYST BEDS AND REACTIVE DISTILLATION FOR THE PRODUCTION OF A LOW SULFUR HEAVY MARINE FUEL OIL

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is: a continuation of U.S. application Ser. No. 16/719,485, filed 18 Dec. 2019; and, U.S. application Ser. No. 16/719,485, is a continuation of U.S. application Ser. No. 16/103,896, filed August 2018, now U.S. Pat. No. 10,836,966; and, U.S. application Ser. No. 16/103,896 is a Continuation-in-part of International Application No. PCT/US2018/017863, filed 12 Feb. 2018; and, International Application No. PCT/US2018/017863 claims priority from U.S. Provisional Application No. 62/589,479, filed 21 Nov. 2017, and, International Application No. PCT/US2018/017863 also claims priority from Provisional Application 62/458,002, filed 12 Feb. 2017, and, U.S. application Ser. No. 16/103,896 is also a Continuation-in-part of International Application No. PCT/US2018/017855, filed 12 Feb. 2018; and, International Application No. PCT/US2018/017855 claims priority from U.S. Provisional Application No. 62/589,479, filed 21 Nov. 2017, and, International Application No. PCT/US2018/017855 also claims priority from Provisional Application 62/458,002, filed 12 Feb. 2017.

BACKGROUND

There are two basic marine fuel types: distillate based marine fuel, also known as Marine Gas Oil (MGO) or Marine Diesel Oil (MDO); and residual based marine fuel, also known as heavy marine fuel oil (HMFO). Distillate based marine fuel both MGO and MDO, comprises petroleum middle distillate fractions separated from crude oil in a refinery via a distillation process. Gasoil (also known as medium diesel) is a petroleum middle distillate in boiling range and viscosity between kerosene (light distillate) and lubricating oil (heavy distillate) containing a mixture of C10 to C19 hydrocarbons. Gasoil (a heavy distillate) is used to heat homes and is used blending with lighter middle distillates as a fuel for heavy equipment such as cranes, bulldozers, generators, bobcats, tractors and combine harvesters. Generally maximizing middle distillate recovery from heavy distillates mixed with petroleum residues is the most economic use of these materials by refiners because they can crack gas oils into valuable gasoline and distillates in a fluid catalytic cracking (FCC) unit. Diesel oils for road use are very similar to gas oils with road use diesel containing predominantly contain a middle distillate mixture of C10 through C19 hydrocarbons, which include approximately 64% aliphatic hydrocarbons, 1-2% olefinic hydrocarbons, and 35% aromatic hydrocarbons. Distillate based marine fuels (MDO and MGO) are essentially road diesel or gas oil fractions blended with up to 15% residual process streams, and optionally up to 5% volume of polycyclic aromatic hydrocarbons (asphaltenes). The residual and asphaltene materials are blended into the middle distillate to form MDO and MGO as a way to both swell volume and productively use these low value materials.

Asphaltenes are large and complex polycyclic hydrocarbons with a propensity to form complex and waxy precipitates, especially in the presence of aliphatic (paraffinic) hydrocarbons that are the primary component of Marine

Diesel. Once asphaltenes have precipitated out, they are notoriously difficult to re-dissolve and are described as fuel tank sludge in the marine shipping industry and marine bunker fueling industry. One of skill in the art will appreciate that mixing Marine Diesel with asphaltenes and process residues is limited by the compatibility of the materials and formation of asphaltene precipitates and the minimum Cetane number required for such fuels.

Residual based fuels or Heavy Marine Fuel Oil (HMFO) are used by large ocean-going ships as fuel for large two stroke diesel engines for over 50 years. HMFO is a blend of the residues generated throughout the crude oil refinery process. Typical refinery streams combined to from HMFO may include, but are not limited to: atmospheric tower bottoms (i.e. atmospheric residues), vacuum tower bottoms (i.e. vacuum residues) visbreaker residue, FCC Light Cycle Oil (LCO), FCC Heavy Cycle Oil (HCO) also known as FCC bottoms, FCC Slurry Oil, heavy gas oils and delayed cracker oil (DCO), deasphalted oils (DAO); heavy aromatic residues and mixtures of polycylic aromatic hydrocarbons, reclaimed land transport motor oils; pyrolysis oils and tars; aspahltene solids and tars; and minor portions (often less than 20% vol.) of middle distillate materials such as cutter oil, kerosene or diesel to achieve a desired viscosity. HMFO has a higher aromatic content (especially polynuclear aromatics and asphaltenes) than the marine distillate fuels noted above. The HMFO component mixture varies widely depending upon the crude slate (i.e. source of crude oil) processed by a refinery and the processes utilized within that refinery to extract the most value out of a barrel of crude oil. The HMFO is generally characterized as being highly viscous, high in sulfur and metal content (up to 5 wt %), and high in asphaltenes making HMFO the one product of the refining process that has historically had a per barrel value less than feedstock crude oil.

Industry statistics indicate that about 90% of the HMFO sold contains 3.5 weight % sulfur. With an estimated total worldwide consumption of HMFO of approximately 300 million tons per year, the annual production of sulfur dioxide by the shipping industry is estimated to be over 21 million tons per year. Emissions from HMFO burning in ships contribute significantly to both global marine air pollution and local marine air pollution levels.

The International Convention for the Prevention of Pollution from Ships, also known as the MARPOL convention or just MARPOL, as administered by the International Maritime Organization (IMO) was enacted to prevent marine pollution (i.e. marpol) from ships. In 1997, a new annex was added to the MARPOL convention; the Regulations for the Prevention of Air Pollution from Ships—Annex VI to minimize airborne emissions from ships (SO_x, NO_x, ODS, VOC) and their contribution to global air pollution. A revised Annex VI with tightened emissions limits was adopted in October 2008 and effective 1 Jul. 2010 (hereafter called Annex VI (revised) or simply Annex VI).

MARPOL Annex VI (revised) adopted in 2008 established a set of stringent air emissions limits for all vessel and designated Emission Control Areas (ECAs). The ECAs under MARPOL Annex VI are: i) Baltic Sea area—as defined in Annex I of MARPOL—SO_x only; ii) North Sea area—as defined in Annex V of MARPOL—SO_x only; iii) North American—as defined in Appendix VII of Annex VI of MARPOL—SO_x, NO_x and PM; and, iv) United States Caribbean Sea area—as defined in Appendix VII of Annex VI of MARPOL—SO_x, NO_x and PM.

Annex VI (revised) was codified in the United States by the Act to Prevent Pollution from Ships (APPS). Under the authority of APPS, the U.S. Environmental Protection Agency (the EPA), in consultation with the United States Coast Guard (USCG), promulgated regulations which incorporate by reference the full text of Annex VI. See 40 C.F.R. § 1043.100(a)(1). On Aug. 1, 2012 the maximum sulfur content of all marine fuel oils used onboard ships operating in US waters/ECA was reduced from 3.5% wt. to 1.00% wt. (10,000 ppm) and on Jan. 1, 2015 the maximum sulfur content of all marine fuel oils used in the North American ECA was lowered to 0.10% wt. (1,000 ppm). At the time of implementation, the United States government indicated that vessel operators must vigorously prepare to comply with the 0.10% wt. (1,000 ppm) US ECA marine fuel oil sulfur standard. To encourage compliance, the EPA and USCG refused to consider the cost of compliant low sulfur fuel oil to be a valid basis for claiming that compliant fuel oil was not available for purchase. For over five years there has been a very strong economic incentive to meet the marine industry demands for low sulfur HMFO, however technically viable solutions have not been realized and a premium price has been commanded by refiners to supply a low sulfur HMFO compliant with Annex VI sulfur emissions requirements in the ECA areas.

Since enactment in 2010, the global sulfur cap for HMFO outside of the ECA areas was set by Annex VI at 3.50% wt. effective 1 Jan. 2012; with a further reduction to 0.50% wt, effective 1 Jan. 2020. The global cap on sulfur content in HMFO has been the subject of much discussion in both the marine shipping and marine fuel bunkering industry. There has been and continues to be a very strong economic incentive to meet the international marine industry demands for low sulfur HMFO (i.e. HMFO with a sulfur content less than 0.50 wt. %. Notwithstanding this global demand, solutions for transforming high sulfur HMFO into low sulfur HMFO have not been realized or brought to market. There is an on-going and urgent demand for processes and methods for making a low sulfur HMFO compliant with MARPOL Annex VI emissions requirements.

Replacement of heavy marine fuel oil with marine gas oil or marine diesel: One primary solution to the demand for low sulfur HMFO to simply replace high sulfur HMFO with marine gas oil (MGO) or marine diesel (MDO). The first major difficulty is the constraint in global supply of middle distillate materials that make up 85-90% vol of MGO and MDO. It is reported that the effective spare capacity to produce MGO is less than 100 million metric tons per year resulting in an annual shortfall in marine fuel of over 200 million metric tons per year. Refiners not only lack the capacity to increase the production of MGO, but they have no economic motivation because higher value and higher margins can be obtained from using middle distillate fractions for low sulfur diesel fuel for land-based transportation systems (i.e. trucks, trains, mass transit systems, heavy construction equipment, etc.).

Blending: Another primary solution is the blending of high sulfur HMFO with lower sulfur containing fuels such as MGO or MDO low sulfur marine diesel (0.1% wt. sulfur) to achieve a Product HMFO with a sulfur content of 0.5% wt. In a straight blending approach (based on linear blending) every 1 ton of high sulfur HSFO (3.5% sulfur) requires 7.5 tons of MGO or MDO material with 0.1% wt. S to achieve a sulfur level of 0.5% wt. HMFO. One of skill in the art of fuel blending will immediately understand that blending hurts key properties of the HMFO, specifically lubricity, fuel density, CCAI, viscosity, flash point and other important physical bulk properties. Blending a mostly paraffinic-type distillate fuel (MGO or MDO) with a HMFO having a high poly aromatic content often correlates with poor solubility of asphaltenes. A blended fuel is likely to result in the precipitation of asphaltenes and/or waxing out of highly paraffinic materials from the distillate material forming an intractable fuel tank sludge. Fuel tank sludge causes clogging of filters and separators, transfer pumps and lines, build-up of sludge in storage tanks, sticking of fuel injection pumps, and plugged fuel nozzles. Such a risk to the primary propulsion system is not acceptable for a ship in the open ocean.

It should further be noted that blending of HMFO with marine distillate products (MGO or MDO) is not economically viable. A blender will be taking a high value product (0.1% S marine gas oil (MGO) or marine diesel (MDO)) and blending it 7.5 to 1 with a low value high sulfur HMFO to create a final IMO/MARPOL compliant HMFO (i.e. 0.5% wt. S Low Sulfur Heavy Marine Fuel Oil—LSHMFO) which will sell at a discount to the value of the principle ingredient (i.e. MGO or MDO).

Processing of residual oils. For the past several decades, the focus of refining industry research efforts related to the processing of heavy oils (crude oils, distressed oils, or residual oils) has been on upgrading the properties of these low value refinery process oils to create middle distillate and lighter oils with greater value. The challenge has been that crude oil, distressed oil and residues contain high levels of sulfur, nitrogen, phosphorous, metals (especially vanadium and nickel); asphaltenes and exhibit a propensity to form carbon or coke on the catalyst. The sulfur and nitrogen molecules are highly refractory and aromatically stable and difficult and expensive to crack or remove. Vanadium and nickel porphyrins and other metal organic compounds are responsible for catalyst contamination and corrosion problems in the refinery. The sulfur, nitrogen, and phosphorous, must be removed because they are well-known poisons for the precious metal (platinum and palladium) catalysts utilized in the processes downstream of the atmospheric or vacuum distillation towers.

The difficulties treating atmospheric or vacuum residual streams has been known for many years and has been the subject of considerable research and investigation. Numerous residue-oil conversion processes have been developed in which the goals are same: 1) create a more valuable, preferably middle distillate range hydrocarbons; and 2) concentrate the contaminates such as sulfur, nitrogen, phosphorous, metals and asphaltenes into a form (coke, heavy coker residue, FCC slurry oil) for removal from the refinery stream. Well known and accepted practice in the refining industry is to increase the reaction severity (elevated temperature and pressure) to produce hydrocarbon products that are lighter and more purified, increase catalyst life times and remove sulfur, nitrogen, phosphorous, metals and asphaltenes from the refinery stream.

In summary, since the announcement of the MARPOL Annex VI standards reducing the global levels of sulfur in HMFO, refiners of crude oil have had modest success in their technical efforts to create a process for the production of a low sulfur substitute for high sulfur HMFO. Despite the strong governmental and economic incentives and needs of the international marine shipping industry, refiners have little economic reason to address the removal of environmental contaminates from high sulfur HMFOs. The global refining industry has been focused upon generating greater value from each barrel of oil by creating middle distillate hydrocarbons (i.e. diesel) and concentrating the environmental contaminates into increasingly lower value streams (i.e. residues) and products (petroleum coke, HMFO). Shipping companies have focused on short term solutions, such as the installation of scrubbing units, or adopting the limited use of more expensive low sulfur marine diesel and marine gas oils as a substitute for HMFO. On the open seas, most if not all major shipping companies continue to utilize the most economically viable fuel, that is HMFO. There remains a long standing and unmet need for processes and devices that remove the environmental contaminants (i.e. sulfur, nitrogen, phosphorous, metals especially vanadium and nickel) from HMFO without altering the qualities and properties that make HMFO the most economic and practical means of powering ocean going vessels.

SUMMARY

It is a general objective to process a high sulfur Feedstock Heavy Marine Fuel Oil (HMFO) in a multi stage process that minimizes the changes in the desirable properties of the HMFO and minimizes the production of by-product hydrocarbons (i.e. light hydrocarbons having C_1-C_4 and wild naphtha (C_5-C_{20})) but reduces the sulfur and other environmental contaminates present in the Feedstock HMFO.

A first aspect and illustrative embodiment encompasses a multi-stage process for the production of a low sulfur Product Heavy Marine Fuel Oil, the process involving: mixing a quantity of the Feedstock Heavy Marine Fuel Oil with a quantity of Activating Gas mixture to give a Feedstock Mixture; contacting the Feedstock Mixture with one or more catalysts in a Reaction System under reactive distillation conditions to form a Process Mixture from the Feedstock Mixture, wherein the one or more catalysts are in a structured catalyst bed; receiving the Process Mixture and separating the Product Heavy Marine Fuel Oil liquid components of the Process Mixture from the gaseous components and by-product hydrocarbon components of the Process Mixture and, discharging the Product Heavy Marine Fuel Oil. In a preferred embodiment, the Reaction System comprises two or more reactor vessel wherein one or more of the reactor vessels contains a structured catalyst bed. In one embodiment, the structured catalyst bed comprises a plurality of catalyst retention structures, each catalyst retention structure composed of a pair of fluid permeable corrugated metal sheets, wherein the pair of the fluid permeable corrugated metal sheets are aligned so the corrugations are out of phase and defining a catalyst rich space and a catalyst lean space within the structured catalyst bed, wherein within the catalyst rich space there is one or more catalyst materials and optionally inert packing materials. Another alternative and embodiment the structured catalyst bed contains one or more catalysts to promote the transformation of the Feedstock mixture to a Process Mixture wherein the structured catalyst bed comprises a plurality of catalyst retention structures, each catalyst retention structure composed of a pair of fluid permeable corrugated metal sheets, wherein the pair of the fluid permeable corrugated metal sheets are aligned so the corrugations are out of phase and defining a catalyst rich space and a catalyst lean space within the structured catalyst bed, wherein within the catalyst rich space there is one or more catalyst materials and optionally inert packing materials.

A second aspect and illustrative embodiment encompasses a device or plant for the production of a Product HMFO. The illustrative devices embody the above illustrative processes on a commercial scale. Such devices include the use of multiple reactor vessels forming a Reactor System in which one or more of the reactor vessels operates under reactive distillation conditions and contains a structured catalyst bed.

A third aspect and illustrative embodiment encompasses a reaction vessel containing one or more structured catalyst beds and configured to operate under reactive distillation conditions so at least 50% and preferably at least 75% and more preferably at least 90% of the desired reaction products are residual in character and exit the lower portion of the reactor vessel.

DESCRIPTION OF DRAWINGS

FIG. **1** is a process block flow diagram of an illustrative core process to produce Product HMFO.

FIG. **2** is a process flow diagram of a multistage process for transforming the Feedstock HMFO and a subsequent core process to produce Product HMFO.

FIG. **3** is a process flow diagram of a first alternative configuration for the Reactor System (**11**) in FIG. **2**.

FIG. **4** is a process flow diagram of a second alternative configuration for the Reactor System (**11**) in FIG. **2**.

FIG. **5** is a side view of a catalyst retention structure component of a first illustrative embodiment of a structured catalyst bed.

FIG. **6** is a side view of a first illustrative embodiment of a structured catalyst bed.

FIG. **7** is a side view of a catalyst retention structure component of a second illustrative embodiment of a structured catalyst bed.

FIG. **8** is a side view of a second illustrative embodiment of a structured catalyst bed.

FIG. **9** illustrates a reaction vessel configured to operate under reactive distillation conditions.

DETAILED DESCRIPTION

The inventive concepts as described herein utilize terms that should be well known to one of skill in the art, however certain terms are utilized having a specific intended meaning and these terms are defined below:

Heavy Marine Fuel Oil (HMFO) is a petroleum product fuel compliant with the ISO 8217 (2017) standards for residual marine fuels except for the concentration levels of the Environmental Contaminates.

Environmental Contaminates are organic and inorganic components of HMFO that result in the formation of SO_x, NO_x and particulate materials upon combustion.

Feedstock HMFO is a petroleum product fuel compliant with the ISO 8217 (2017) standards for the physical properties or characteristics of a merchantable HMFO except for the concentration of Environmental Contaminates, more specifically the Feedstock HMFO has a sulfur content greater than the global MARPOL Annex VI standard of 0.5% wt. sulfur, and preferably and has a sulfur content (ISO 14596 or ISO 8754) between the range of 5.0% wt. to 1.0% wt.

Product HMFO is a petroleum product fuel compliant with the ISO 8217 (2017) standards for the physical properties or characteristics of a merchantable HMFO and has a sulfur content lower than the global MARPOL Annex VI standard of 0.5% wt. sulfur (ISO 14596 or ISO 8754), and preferably a maximum sulfur content (ISO 14596 or ISO 8754) between the range of 0.05% wt. to 1.0% wt.

Activating Gas: is a mixture of gases utilized in the process combined with the catalyst to remove the environmental contaminates from the Feedstock HMFO.

Fluid communication: is the capability to transfer fluids (either liquid, gas or combinations thereof, which might have suspended solids) from a first vessel or location to a second vessel or location, this may encompass connections made by pipes (also called a line), spools, valves, intermediate holding tanks or surge tanks (also called a drum).

Merchantable quality: is a level of quality for a residual marine fuel oil so the fuel is fit for the ordinary purpose it should serve (i.e. serve as a residual fuel source for a marine ship) and can be commercially sold as and is fungible and compatible with other heavy or residual marine bunker fuels.

Bbl or bbl: is a standard volumetric measure for oil; 1 bbl=0.1589873 m^3; or 1bbl=158.9873 liters; or 1 bbl=42.00 US liquid gallons.

Bpd or bpd: is an abbreviation for Bbl per day.

SCF: is an abbreviation for standard cubic foot of a gas; a standard cubic foot (at 14.73 psi and 60° F.) equals 0.0283058557 standard cubic meters (at 101.325 kPa and 15° C.).

Bulk Properties: are broadly defined as the physical properties or characteristics of a merchantable MFO as required by ISO 8217 (2017); and more specifically the measurements include: kinematic viscosity at 50° C. as determined by ISO 3104; density at 15° C. as determined by ISO 3675; CCAI value as determined by ISO 8217, ANNEX B; flash point as determined by ISO 2719; total sediment—aged as determined by ISO 10307-2; carbon residue—micro method as determined by ISO 10370; and also preferably aluminum plus silicon content as determined by ISO 10478.

The inventive concepts are illustrated in more detail in this description referring to the drawings, in which FIG. **1** shows the generalized block process flows for a core process of reducing the Environmental Contaminates in a Feedstock MFO and producing a Product HMFO. A predetermined volume of Feedstock HMFO (2) is mixed with a predetermined quantity of Activating Gas (4) to give a Feedstock Mixture. The Feedstock HMFO utilized generally complies with the bulk physical and certain key chemical properties for a residual marine fuel oil otherwise compliant with ISO 8217 (2017) exclusive of the Environmental Contaminates. More particularly, when the Environmental Contaminate is sulfur, the concentration of sulfur in the Feedstock HMFO may be between the range of 5.0% wt. to 1.0% wt. The Feedstock HMFO should have bulk physical properties required of an ISO 8217 (2017) compliant HMFO. The Feedstock HMFO should exhibit the Bulk Properties of: a maximum of kinematic viscosity at 50° C. (ISO 3104) between the range from 180 mm^2/s to 700 mm^2/s; a maximum of density at 15° C. (ISO 3675) between the range of 991.0 kg/m^3 to 1010.0 kg/m^3; a CCAI in the range of 780 to 870; and a flash point (ISO 2719) no lower than 60° C. Properties of the Feedstock HMFO connected to the formation of particulate material (PM) include: a total sediment—aged (ISO 10307-2) less than 0.10% wt. and a carbon residue—micro method (ISO 10370) less than 20% wt. and preferably a aluminum plus silicon (ISO 10478) content of less than 60 mg/kg. Environmental Contaminates other than sulfur that may be present in the Feedstock HMFO over the ISO requirements may include vanadium, nickel, iron, aluminum and silicon substantially reduced by the process of the present invention. However, one of skill in the art will appreciate that the vanadium content serves as a general indicator of these other Environmental Contaminates. In one preferred embodiment the vanadium content is ISO compliant so the Feedstock HMFO has a maximum vanadium content (ISO 14597) between the range from 350 mg/kg to 450 ppm mg/kg.

As for the properties of the Activating Gas, the Activating Gas should be selected from mixtures of nitrogen, hydrogen, carbon dioxide, gaseous water, and methane. The mixture of gases within the Activating Gas should have an ideal gas partial pressure of hydrogen (PH2) greater than 90% of the total pressure of the Activating Gas mixture (P) and more preferably wherein the Activating Gas has an ideal gas partial pressure of hydrogen (p_{H2}) greater than 95% of the total pressure of the Activating Gas mixture (P). It will be appreciated by one of skill in the art that the molar content of the Activating Gas is another criterion the Activating Gas should have a hydrogen mole fraction in the range between 80% and 100% of the total moles of Activating Gas mixture.

The Feedstock Mixture (i.e. mixture of Feedstock HMFO and Activating Gas) is brought up to the process reactive conditions of temperature and pressure and introduced into a Reactor System, preferably a reactor vessel, so the Feedstock Mixture is then contacted under reactive conditions with one or more catalysts (8) to form a Process Mixture from the Feedstock Mixture.

The core process reactive conditions are selected so the ratio of the quantity of the Activating Gas to the quantity of Feedstock HMFO is 250 scf gas/bbl of Feedstock HMFO to 10,000 scf gas/bbl of Feedstock HMFO; and preferably between 2000 scf gas/bbl of Feedstock HMFO 1 to 5000 scf gas/bbl of Feedstock HMFO more preferably between 2500 scf gas/bbl of Feedstock HMFO to 4500 scf gas/bbl of Feedstock HMFO. The process conditions are selected so the total pressure in the first vessel is between of 250 psig and 3000 psig; preferably between 1000 psig and 2500 psig, and more preferably between 1500 psig and 2200 psig. The process reactive conditions are selected so the indicated temperature within the first vessel is between of 500° F. to 900° F., preferably between 650° F. and 850° F. and more preferably between 680° F. and 800° F. The process conditions are selected so the liquid hourly space velocity within the first vessel is between 0.05 oil/hour/m^3 catalyst and 1.0 oil/hour/m^3 catalyst; preferably between 0.08 oil/hour/m^3 catalyst and 0.5 oil/hour/m^3 catalyst; and more preferably between 0.1 oil/hour/m^3 catalyst and 0.3 oil/hour/m^3 catalyst to achieve deep desulfurization with product sulfur levels below 0.1 ppmw.

One of skill in the art will appreciate that the core process reactive conditions are determined considering the hydraulic capacity of the unit. Exemplary hydraulic capacity for the treatment unit may be between 100 bbl of Feedstock HMFO/day and 100,000 bbl of Feedstock HMFO/day, preferably between 1000 bbl of Feedstock HMFO/day and 60,000 bbl of Feedstock HMFO/day, more preferably between 5,000 bbl of Feedstock HMFO/day and 45,000 bbl of Feedstock HMFO/day, and even more preferably between 10,000 bbl of Feedstock HMFO/day and 30,000 bbl of Feedstock HMFO/day.

The transition metal heterogeneous catalyst utilized comprises a porous inorganic oxide catalyst carrier and a transition metal catalytic metal. The porous inorganic oxide catalyst carrier is at least one carrier selected from the group consisting of alumina, alumina/boria carrier, a carrier containing metal-containing aluminosilicate, alumina/phosphorus carrier, alumina/alkaline earth metal compound carrier, alumina/titania carrier and alumina/zirconia carrier. The transition metal catalytic metal component of the catalyst is one or more metals selected from the group consisting of group 6, 8, 9 and 10 of the Periodic Table. In a preferred and illustrative embodiment, the transition metal heterogeneous catalyst is a porous inorganic oxide catalyst carrier and a transition metal catalyst, in which the preferred porous inorganic oxide catalyst carrier is alumina and the preferred transition metal catalyst is Ni—Mo, Co—Mo, Ni—W or

Ni—Co—Mo. The process by which the transition metal heterogeneous catalyst is manufactured is known in the literature and preferably the catalysts are commercially available as hydrodemetallization catalysts, transition catalysts, desulfurization catalyst and combinations of these which might be pre-sulfided. The transition metal heterogeneous catalyst is part of a structured catalyst bed in which the catalyst particles are distributed within an otherwise inert supporting structure to achieve an optimized balance between catalyst activity per unit volume and the linear space velocity of the Feedstock Mixture. This is described in greater detail below in relation to the structured catalyst bed configuration itself.

The Process Mixture (**10**) in this core process is removed from the Reactor System (**8**) and from being in contact with the one or more catalyst and is sent via fluid communication to a second vessel (**12**), preferably a gas-liquid separator or hot separators and cold separators, for separating the liquid components (**14**) of the Process Mixture from the bulk gaseous components (**16**) of the Process Mixture. The gaseous components (**16**) are treated beyond the battery limits of the immediate process. Such gaseous components may include a mixture of Activating Gas components and lighter hydrocarbons (mostly methane, ethane and propane but some wild naphtha) that may have been formed as part of the by-product hydrocarbons from the process.

The Liquid Components (**16**) in this core process are sent via fluid communication to a third vessel (**18**), preferably a fuel oil product stripper system, for separating any residual gaseous components (**20**) and by-product hydrocarbon components (**22**) from the Product HMFO (**24**). The residual gaseous components (**20**) may be a mixture of gases selected from the group consisting of: nitrogen, hydrogen, carbon dioxide, hydrogen sulfide, gaseous water, C_1-C_5 hydrocarbons. This residual gas is treated outside of the battery limits of the immediate process, combined with other gaseous components (**16**) removed from the Process Mixture (**10**) in the second vessel (**12**). The liquid by-product hydrocarbon component, which are condensable hydrocarbons formed in the process (**22**) may be a mixture selected from the group consisting of C_5-C_{20} hydrocarbons (wild naphtha) (naphtha-diesel) and other condensable light liquid (C_3-C_5) hydrocarbons that can be utilized as part of the motor fuel blending pool or sold as gasoline and diesel blending components on the open market. We also assume a second draw (not shown) may be included to withdraw a distillate product, preferably a middle to heavy distillate. These liquid by-product hydrocarbons should be less than 15% wt., preferably less than 5% wt. and more preferably less than 3% wt. of the overall process mass balance.

The Product HMFO (**24**) resulting from the core process is discharged via fluid communication into storage tanks beyond the battery limits of the immediate process. The Product HMFO complies with ISO 8217 (2017) and has a maximum sulfur content (ISO 14596 or ISO 8754) between the range of 0.05 mass % to 1.0 mass % preferably a sulfur content (ISO 14596 or ISO 8754) between the range of 0.05 mass % ppm and 0.7 mass % and more preferably a sulfur content (ISO 14596 or ISO 8754) between the range of 0.1 mass % and 0.5 mass %. The vanadium content of the Product HMFO is also ISO compliant with a maximum vanadium content (ISO 14597) between the range from 350 mg/kg to 450 ppm mg/kg, preferably a vanadium content (ISO 14597) between the range of 200 mg/kg and 300 mg/kg and more preferably a vanadium content (ISO 14597) between the range of 50 mg/kg and 100 mg/kg.

The Product HFMO should have bulk physical properties that are ISO 8217 (2017) compliant. The Product HMFO should exhibit Bulk Properties of: a maximum of kinematic viscosity at 50° C. (ISO 3104) between the range from 180 mm^2/s to 700 mm^2/s; a maximum of density at 15° C. (ISO 3675) between the range of 991.0 kg/m^3 to 1010.0 kg/m^3; a CCAI value in the range of 780 to 870; a flash point (ISO 2719) no lower than 60.0° C.; a total sediment—aged (ISO 10307-2) of less than 0.10 mass %; a carbon residue—micro method (ISO 10370) lower than 20 mass %, and also preferably an aluminum plus silicon (ISO 10478) content of less than 60 mg/kg.

Relative the Feedstock HMFO, the Product HMFO will have a sulfur content (ISO 14596 or ISO 8754) between 1% and 20% of the maximum sulfur content of the Feedstock HMFO. That is the sulfur content of the Product will be reduced by about 80% or greater when compared to the Feedstock HMFO. Similarly, the vanadium content (ISO 14597) of the Product HMFO is between 1% and 20% of the maximum vanadium content of the Feedstock HMFO. One of skill in the art will appreciate that the above data indicates a substantial reduction in sulfur and vanadium content indicate a process having achieved a substantial reduction in the Environmental Contaminates from the Feedstock HMFO while maintaining the desirable properties of an ISO 8217 compliant and merchantable HMFO.

As a side note, the residual gaseous component is a mixture of gases selected from the group consisting of: nitrogen, hydrogen, carbon dioxide, hydrogen sulfide, gaseous water, C_1-C_5 hydrocarbons. An amine scrubber will effectively remove the hydrogen sulfide content which can then be processed using technologies and processes well known to one of skill in the art. In one preferable illustrative embodiment, the hydrogen sulfide is converted into elemental sulfur using the well-known Claus process. An alternative embodiment utilizes a proprietary process for conversion of the Hydrogen sulfide to hydrosulfuric acid. Either way, the sulfur is removed from entering the environment prior to combusting the HMFO in a ships engine. The cleaned gas can be vented, flared or more preferably recycled back for use as Activating Gas.

Product HMFO The Product HFMO resulting from the disclosed illustrative process is of merchantable quality for sale and use as a heavy marine fuel oil (also known as a residual marine fuel oil or heavy bunker fuel) and exhibits the bulk physical properties required for the Product HMFO to be an ISO compliant (i.e. ISO 8217 (2017)) residual marine fuel oil. The Product HMFO should exhibit the Bulk Properties of: a maximum of kinematic viscosity at 50° C. (ISO 3104) between the range from 180 mm^2/s to 700 mm^2/s; a density at 15° C. (ISO 3675) between the range of 991.0 kg/m^3 to 1010.0 kg/m^3; a CCAI is in the range of 780 to 870; a flash point (ISO 2719) no lower than 60° C.; a total sediment—aged (ISO 10307-2) less than 0.10% wt.; and a carbon residue—micro method (ISO 10370) less than 20% wt.; and also preferably an aluminum plus silicon (ISO 10478) content no more than of 60 mg/kg.

The Product HMFO has a sulfur content (ISO 14596 or ISO 8754) less than 0.5 wt % and preferably less than 0.1% wt. and complies with the IMO Annex VI (revised) requirements for a low sulfur and preferably an ultra-low sulfur HMFO. That is the sulfur content of the Product HMFO has been reduced by about 80% or greater when compared to the Feedstock HMFO. Similarly, the vanadium content (ISO 14597) of the Product Heavy Marine Fuel Oil is less than 20% and more preferably less than 10% of the maximum vanadium content of the Feedstock Heavy Marine Fuel Oil.

One of skill in the art will appreciate that a substantial reduction in sulfur and vanadium content of the Feedstock HMFO indicates a process having achieved a substantial reduction in the Environmental Contaminates from the Feedstock HMFO; of equal importance is this has been achieved while maintaining the desirable properties of an ISO 8217 (2017) compliant HMFO.

The Product HMFO not only complies with ISO 8217 (2017) (and is merchantable as a residual marine fuel oil or bunker fuel), the Product HMFO has a maximum sulfur content (ISO 14596 or ISO 8754) between the range of 0.05% wt. to 1.0% wt. preferably a sulfur content (ISO 14596 or ISO 8754) between the range of 0.05% wt. ppm and 0.5% wt. and more preferably a sulfur content (ISO 14596 or ISO 8754) between the range of 0.1% wt. and 0.5% wt. The vanadium content of the Product HMFO is well within the maximum vanadium content (ISO 14597) required for an ISO 8217 (2017) residual marine fuel oil exhibiting a vanadium content lower than 450 ppm mg/kg, preferably a vanadium content (ISO 14597) lower than 300 mg/kg and more preferably a vanadium content (ISO 14597) less than 50 mg/kg.

One knowledgeable in the art of marine fuel blending, bunker fuel formulations and the fuel requirements for marine shipping fuels will readily appreciate that without further compositional changes or blending, the Product HMFO can be sold and used as a low sulfur MARPOL Annex VI compliant heavy (residual) marine fuel oil that is a direct substitute for the high sulfur heavy (residual) marine fuel oil or heavy bunker fuel currently in use. One illustrative embodiment is an ISO 8217 (2017) compliant low sulfur heavy marine fuel oil comprising (and preferably consisting essentially of) hydroprocessed ISO 8217 (2017) compliant high sulfur heavy marine fuel oil, wherein the sulfur levels of the hydroprocessed ISO 8217 (2017) compliant high sulfur heavy marine fuel oil is greater than 0.5% wt. and wherein the sulfur levels of the ISO 8217 (2017) compliant low sulfur heavy marine fuel oil is less than 0.5% wt. Another illustrative embodiment is an ISO 8217 (2017) compliant ultra-low sulfur heavy marine fuel oil comprising (and preferably consisting essentially of) a hydroprocessed ISO 8217 (2017) compliant high sulfur heavy marine fuel oil, wherein the sulfur levels of the hydroprocessed ISO 8217 (2017) compliant high sulfur heavy marine fuel oil is greater than 0.5% wt. and wherein the sulfur levels of the ISO 8217 (2017) compliant low sulfur heavy marine fuel oil is less than 0.1% wt.

Because of the present invention, multiple economic and logistical benefits to the bunkering and marine shipping industries can be realized. The benefits include minimal changes to the existing heavy marine fuel bunkering infrastructure (storage and transferring systems); minimal changes to shipboard systems are needed to comply with emissions requirements of MARPOL Annex VI (revised); no additional training or certifications for crew members will be needed, amongst the realizable benefits. Refiners will also realize multiple economic and logistical benefits, including: no need to alter or rebalance the refinery operations, crude slate, and product streams to meet a new market demand for low sulfur or ultralow sulfur HMFO; no additional units are needed in the refinery with additional hydrogen or sulfur capacity because the illustrative process can be conducted as a stand-alone unit; refinery operations can remain focused on those products that create the greatest value from the crude oil received (i.e. production of petrochemicals, gasoline and distillate (diesel); refiners can continue using the existing slates of crude oils without having to switch to sweeter or lighter crudes to meet the environmental requirements for HMFO products.

Heavy Marine Fuel Composition One aspect of the present inventive concept is a fuel composition comprising, but preferably consisting essentially of, the Product HMFO resulting from the processes disclosed, and may optionally include Diluent Materials. The Product HMFO itself complies with ISO 8217 (2017) and meets the global IMO Annex VI requirements for maximum sulfur content (ISO 14596 or ISO 8754). If ultra-low levels of sulfur are desired, the process of the present invention achieves this and one of skill in the art of marine fuel blending will appreciate that a low sulfur or ultra-low sulfur Product HMFO can be utilized as a primary blending stock to form a global IMO Annex VI compliant low sulfur Heavy Marine Fuel Composition. Such a low sulfur Heavy Marine Fuel Composition will comprise (and preferably consist essentially of): a) the Product HMFO and b) Diluent Materials. In one embodiment, the majority of the volume of the Heavy Marine Fuel Composition is the Product HMFO with the balance of materials being Diluent Materials. Preferably, the Heavy Maine Fuel Composition is at least 75% by volume, preferably at least 80% by volume, more preferably at least 90% by volume, and furthermore preferably at least 95% by volume Product HMFO with the balance being Diluent Materials.

Diluent Materials may be hydrocarbon or non-hydrocarbon based materials mixed into or combined with or added to, or solid particle materials suspended in, the Product HMFO. The Diluent Materials may intentionally or unintentionally alter the composition of the Product HMFO but not so the resulting mixture violates the ISO 8217 (2017) standards for residual marine fuels or fails to have a sulfur content lower than the global MARPOL standard of 0.5% wt. sulfur (ISO 14596 or ISO 8754). Examples of Diluent Materials considered hydrocarbon based materials include: Feedstock HMFO (i.e. high sulfur HMFO); distillate based fuels such as road diesel, gas oil, MGO or MDO; cutter oil (which is used in formulating residual marine fuel oils); renewable oils and fuels such as biodiesel, methanol, ethanol, and the like; synthetic hydrocarbons and oils based on gas to liquids technology such as Fischer-Tropsch derived oils, synthetic oils such as those based on polyethylene, polypropylene, dimer, trimer and poly butylene; refinery residues or other hydrocarbon oils such as atmospheric residue, vacuum residue, fluid catalytic cracker (FCC) slurry oil, FCC cycle oil, pyrolysis gasoil, cracked light gas oil (CLGO), cracked heavy gas oil (CHGO), light cycle oil (LCO), heavy cycle oil (HCO), thermally cracked residue, coker heavy distillate, bitumen, de-asphalted heavy oil, visbreaker residue, slop oils, asphaltinic oils; used or recycled motor oils; lube oil aromatic extracts and crude oils such as heavy crude oil, distressed crude oils and similar materials that might otherwise be sent to a hydrocracker or diverted into the blending pool for a prior art high sulfur heavy (residual) marine fuel oil. Examples of Diluent Materials considered non-hydrocarbon based materials include: residual water (i.e. water absorbed from the humidity in the air or water that is miscible or solubilized, sometimes as microemulsions, into the hydrocarbons of the Product HMFO), fuel additives which can include, but are not limited to detergents, viscosity modifiers, pour point depressants, lubricity modifiers, de-hazers (e.g. alkoxylated phenol formaldehyde polymers), antifoaming agents (e.g. polyether modified polysiloxanes); ignition improvers; anti rust agents (e.g. succinic acid ester derivatives); corrosion inhibitors; anti-wear additives, anti-oxidants (e.g. phenolic compounds and derivatives), coating agents and surface modifiers, metal deactivators, static dissipating agents, ionic and nonionic surfactants, stabilizers, cosmetic colorants and odorants and mixtures of these. A third group of Diluent Materials may include suspended solids or fine particulate materials that are present because of the handling, storage and transport of the Product HMFO or the Heavy Marine Fuel Composition, including but not limited to: carbon or hydrocarbon solids (e.g. coke, graphitic solids, or micro-agglomerated asphaltenes), iron rust and other oxidative corrosion solids, fine bulk metal particles, paint or surface coating particles, plastic or polymeric or elastomer or rubber particles (e.g. resulting from the degradation of gaskets, valve parts, etc.,), catalyst fines, ceramic or mineral particles, sand, clay, and other earthen particles, bacteria and other biologically generated solids, and mixtures of these that may be present as suspended particles, but otherwise don't detract from the merchantable quality of the Heavy Marine Fuel Composition as an ISO 8217 (2017) compliant heavy (residual) marine fuel.

The blend of Product HMFO and Diluent Materials must be of merchantable quality as a low sulfur heavy (residual) marine fuel. That is the blend must be suitable for the intended use as heavy marine bunker fuel and generally be fungible and compatible as a bunker fuel for ocean going ships. Preferably the Heavy Marine Fuel Composition must retain the bulk physical properties required of an ISO 8217 (2017) compliant residual marine fuel oil and a sulfur content lower than the global MARPOL standard of 0.5% wt. sulfur (ISO 14596 or ISO 8754) so that the material qualifies as MARPOL Annex VI Low Sulfur Heavy Marine Fuel Oil (LS-HMFO). The sulfur content of the Product HMFO can be lower than 0.5% wt. (i.e. below 0.1% wt sulfur (ISO 14596 or ISO 8754)) to qualify as a MARPOL Annex VI compliant Ultra-Low Sulfur Heavy Marine Fuel Oil (ULS-HMFO) and a Heavy Marine Fuel Composition likewise can be formulated to qualify as a MARPOL Annex VI compliant ULS-HMFO suitable for use as marine bunker fuel in the ECA zones. To qualify as an ISO 8217 (2017) qualified fuel, the Heavy Marine Fuel Composition of the present invention must meet those internationally accepted standards. The Bulk Properties of the Heavy Marine Fuel Composition should include: a maximum of kinematic viscosity at 50° C. (ISO 3104) between the range from 180 mm^2/s to 700 mm^2/s; a density at 15° C. (ISO 3675) between the range of 991.0 kg/m^3 to 1010.0 kg/m^3; a CCAI is in the range of 780 to 870; a flash point (ISO 2719) no lower than 60° C.; a total sediment—aged (ISO 10307-2) less than 0.10% wt.; and a carbon residue—micro method (ISO 10370) less than 20% wt., and an aluminum plus silicon (ISO 10478) content no more than of 60 mg/kg.

Production Plant Description Production Plant Description: Turning now to a more detailed illustrative embodiment of a production plant, FIG. **2** shows a schematic for a production plant implementing the process described above for reducing the environmental contaminates in a Feedstock HMFO to produce a Product HMFO according to the second illustrative embodiment. One of skill in the art will appreciate that FIG. **2** is a generalized schematic drawing, and the exact layout and configuration of a plant will depend upon factors such as location, production capacity, environmental conditions (i.e. wind load, etc.) and other factors and elements that a skilled detailed engineering firm can provide. Such variations are contemplated and within the scope of the present disclosure.

In FIG. **2**, Feedstock HMFO (A) is fed from outside the battery limits (OSBL) to the Oil Feed Surge Drum (**1**) that receives feed from outside the battery limits (OSBL) and provides surge volume adequate to ensure smooth operation of the unit. Water entrained in the feed and bulk solids (sand, rust particles, etc.) are removed from the HMFO with the water and bulk solids being discharged a stream (**1**c) for treatment OSBL.

The Feedstock HMFO (A) is withdrawn from the Oil Feed Surge Drum (**1**) via line (**1**b) by the Oil Feed Pump (**3**) and is pressurized to a pressure required for the process. The pressurized HMFO (A') then passes through line (**3**a) to the Oil Feed/Product Heat Exchanger (**5**) where the pressurized HMFO Feed (A') is partially heated by the Product HMFO (B). The pressurized Feedstock HMFO (A') passing through line (**5**a) is further heated against the effluent from the Reactor System (E) in the Reactor Feed/Effluent Heat Exchanger (**7**).

The heated and pressurized Feedstock HMFO (A") in line (**7**a) is then mixed with Activating Gas (C) provided via line (**23**c) at Mixing Point (X) to form a Feedstock Mixture (D). The mixing point (X) can be any well know gas/liquid mixing system or entrainment mechanism well known to one skilled in the art.

The Feedstock Mixture (D) passes through line (**9**a) to the Reactor Feed Furnace (**9**) where the Feedstock Mixture (D) is heated to the specified process temperature. The Reactor Feed Furnace (**9**) may be a fired heater furnace or any other kind to type of heater as known to one of skill in the art if it will raise the temperature of the Feedstock Mixture (D) to the desired temperature for the process conditions.

The fully Heated Feedstock Mixture (D') exits the Reactor Feed Furnace (**9**) via line **9**b and is fed into the Reactor System (**11**). The fully heated Feedstock Mixture (D') enters the Reactor System (**11**) where environmental contaminates, such a sulfur, nitrogen, and metals are preferentially removed from the Feedstock HMFO component of the fully heated Feedstock Mixture. The Reactor System contains a catalyst materials some or all of which many be within a structured catalyst bed, and which preferentially removes the Environmental Contaminates, principally organometal and organosulfur compounds in the Feedstock HMFO component by reacting them with hydrogen in the Activating Gas to form deposits of metals, metal sulfides and/or hydrogen sulfide. The Reactor System will also achieve demetallization, denitrogenation, and a certain amount of ring opening hydrogenation of the complex aromatics and asphaltenes, however minimal hydrocracking of hydrocarbons should take place. The process conditions of hydrogen partial pressure, reaction pressure, temperature and residence time as measured by liquid hourly velocity are optimized to achieve desired final product quality. A more detailed discussion of the Reactor System, the structured catalyst beds, the catalyst, the process conditions, and other aspects of the process are contained below in the "Reactor System Description."

The Reactor System Effluent (E) exits the Reactor System (**11**) via line (**11**a) and exchanges heat against the pressurized and partially heats the Feedstock HMFO (A') in the Reactor Feed/Effluent Exchanger (**7**). The partially cooled Reactor System Effluent (E') then flows via line (**11**c) to the Hot Separator (**13**).

The Hot Separator (**13**) separates the gaseous components of the Reactor System Effluent (F) which are directed to line (**13**a) from the liquid components of the Reactor System effluent (G) which are directed to line (**13**b). The gaseous components of the Reactor System effluent in line (**13**a) are cooled against air in the Hot Separator Vapor Air Cooler (**15**) and then flow via line (**15**a) to the Cold Separator (**17**).

The Cold Separator (**17**) further separates any remaining gaseous components from the liquid components in the cooled gaseous components of the Reactor System Effluent (F'). The gaseous components from the Cold Separator (F") are directed to line (**17***a*) and fed onto the Amine Absorber (**21**). The Cold Separator (**17**) also separates any remaining Cold Separator hydrocarbon liquids (H) in line (**17***b*) from any Cold Separator condensed liquid water (I). The Cold Separator condensed liquid water (I) is sent OSBL via line (**17***c*) for treatment.

The hydrocarbon liquid components of the Reactor System effluent from the Hot Separator (G) in line (**13***b*) and the Cold Separator hydrocarbon liquids (H) in line (**17***b*) are combined and are fed to the Oil Product Stripper System (**19**). The Oil Product Stripper System (**19**) removes any residual hydrogen and hydrogen sulfide from the Product HMFO (B) which is discharged in line (**19***b*) to storage OSBL. We also assume a second draw (not shown) may be included to withdraw a distillate product, preferably a middle to heavy distillate. The vent stream (M) from the Oil Product Stripper in line (**19***a*) may be sent to the fuel gas system or to the flare system that are OSBL. A more detailed discussion of the Oil Product Stripper System is contained in the "Oil Product Stripper System Description."

The gaseous components from the Cold Separator (F") in line (**17***a*) contain a mixture of hydrogen, hydrogen sulfide and light hydrocarbons (mostly methane and ethane). This vapor stream (**17***a*) feeds an Amine Absorber System (**21**) where it is contacted against Lean Amine (J) provided OSBL via line (**21***a*) to the Amine Absorber System (**21**) to remove hydrogen sulfide from the gases making up the Activating Gas recycle stream (C'). Rich amine (K) which has absorbed hydrogen sulfide exits the bottom of the Amine Absorber System (**21**) and is sent OSBL via line (**21***b*) for amine regeneration and sulfur recovery.

The Amine Absorber System overhead vapor in line (**21***c*) is preferably recycled to the process as a Recycle Activating Gas (C') via the Recycle Compressor (**23**) and line (**23***a*) where it is mixed with the Makeup Activating Gas (C") provided OSBL by line (**23***b*). This mixture of Recycle Activating Gas (C') and Makeup Activating Gas (C") to form the Activating Gas (C) utilized in the process via line (**23***c*) as noted above. A Scrubbed Purge Gas stream (H) is taken from the Amine Absorber System overhead vapor line (**21***c*) and sent via line (**21***d*) to OSBL to prevent the buildup of light hydrocarbons or other non-condensable hydrocarbons. A more detailed discussion of the Amine Absorber System is contained in the "Amine Absorber System Description."

Reactor System Description: The core process Reactor System (**11**) illustrated in FIG. **2** comprises a single reactor vessel loaded with the process catalyst and sufficient controls, valves and sensors as one of skill in the art would readily appreciate. One of skill in the art will appreciate that the reactor vessel itself must be engineered to withstand the pressures, temperatures and other conditions (i.e. presence of hydrogen and hydrogen sulfide) of the process. Using special alloys of stainless steel and other materials typical of such a unit are within the skill of one in the art and well known. As illustrated, one or more beds in the reactor vessel may be a structured catalyst bed or a fixed dense packed catalyst bed or combinations of these. A Reactor System containing at least reactor vessel with at least one structured catalyst bed is preferred as these are easier to operate and maintain, however other reactor types are also within the scope of the invention.

A description of the process catalyst, the selection of the process catalyst and the grading of the catalyst within the reactor vessel is contained in the "Catalyst in Reactor System".

Alternative configurations for the core process Reactor System (**11**) are contemplated. In one illustrative configuration, more than one reactor vessel may be utilized in parallel as shown in FIG. **3** to replace the core process Reactor System (**11**) illustrated in FIG. **2**.

In the embodiment in FIG. **3**, each reactor vessel is loaded with process catalyst in a similar manner and each reactor vessel in the Reactor System is provided the heated Feed Mixture (D') via a common line **9***b*. The effluent from each reactor vessel in the Reactor System is recombined and forms a combined Reactor Effluent (E) for further processing as described above via line **11***a*. The illustrated arrangement will allow the three reactors to carry out the process effectively multiplying the hydraulic capacity of the overall Reactor System. Control valves and isolation valves may also prevent feed from entering one reactor vessel but not another reactor vessel. In this way one reactor can be by-passed and placed off-line for maintenance and reloading of catalyst while the remaining reactors continues to receive heated Feedstock Mixture (D'). It will be appreciated by one of skill in the art this arrangement of reactor vessels in parallel is not limited in number to three, but multiple additional reactor vessels can be added as shown by dashed line reactor. The only limitation to the number of parallel reactor vessels is plot spacing and the ability to provide heated Feedstock Mixture (D') to each active reactor.

A cascading series in FIG. **4** can also be substituted for the single reactor vessel Reactor System **11** in FIG. **2**. The cascading reactor vessels are loaded with process catalyst with the same or different activities toward metals, sulfur or other environmental contaminates to be removed. For example, one reactor may be loaded with a highly active demetallization catalyst, a second subsequent or downstream reactor may be loaded with a balanced demetallization/desulfurizing catalyst, and reactor downstream from the second reactor may be loaded with a highly active desulfurization catalyst. This allows for greater control and balance in process conditions (temperature, pressure, space flow velocity, etc.) so it is tailored for each catalyst. In this way one can optimize the parameters in each reactor depending upon the material being fed to that specific reactor/catalyst combination and minimize the hydrocracking reactions.

Another illustrative embodiment of the replacement of the single reactor vessel Reactor System **11** in FIG. **2** is a matrix of reactors composed of interconnected reactors in parallel and in series. A simple 2×2 matrix arrangement of reactors with associated control valves and piping illustrates this embodiment, however a wide variety of matrix configurations such as 2×3; 3×3, etc., are contemplated and within the scope of the present invention. A 2 reactor by 2 reactor (2×2) matrix of comprises four reactor vessels each with reactor inlet control valves and reactor outlet control valves associated with each vessel. Horizontal flow control valves regulate the flow across the matrix from heated Feedstock (D') in line **9***b* to discharging Reactor Effluent (E) into line **11***a*. Vertical flow control valves control the flow through the matrix from line **9***b* to line **11***a*. One of skill in the art will appreciate that control over the multiple valves and flow can be achieved using a computerized control system/distributed control system (DCS) or programmable logic controllers (PLC) programed to work with automatic motorized valve controls, position sensors, flow meters, thermocouples, etc. These systems are commercially available from vendors such as Honeywell International, Schneider Electric; and ABB. Such control systems will include lock-outs and other process safety control systems to prevent opening of valves in manner either not productive or unsafe.

One of skill in the art will quickly realize and appreciate that by opening and closing the valves and varying the catalyst loads present in each reactor, many configurations may be achieved. One such configuration would be to open valves and close other valves so the flow for heated Feed Mixture (D') will pass through the reactors in series. Another such configuration would be to open valves and close other valves so the flow of heated Feed Mixture (D') will pass through three of the reactors in series, but not the fourth. As with the prior example, the nature of the Feedstock HSFO and the catalyst loaded in each reactor and the presence or absence of structured catalyst beds may be optimized and adjusted to achieve the desired Product HSFO properties, however for brevity of disclose all such variations will be apparent to one of skill in the art.

One benefit of having a multi-reactor Reactor System is that it allows for a reactor that is experiencing decreased activity or plugging because of coke formation to be isolated and taken off line for turn-around (i.e. deactivated, catalyst and internals replaced, etc.) without the entire plant having to shut down. Another benefit as noted above is that it allows one to vary the catalyst loading in the Reactor System so the overall process can be optimized for a specific Feedstock HSFO. A further benefit is that one can design the piping, pumps, heaters/heat exchangers, etc. to have excess capacity so that when an increase in capacity is desired, additional reactors can be quickly brought on-line. Conversely, it allows an operator to take capacity off line, or turn down a plant output without having a concern about turn down and minimum flow through a reactor. While the above matrix Reactor System is described referring to a fixed bed or packed bed trickle flow reactor, one of skill in the art will appreciate that other reactor types may be utilized. For example, one or more reactors may be configured to be ebullated bed up flow reactors or three phase upflow bubble reactors, or counter-current reactors, the configuration of which will be known to one of skill in the art. In a preferred embodiment, at least one reactor is configured to operate using structured catalyst beds and under reactive distillation configuration of the type disclosed. It is anticipated that many other operational and logistical benefits will be realized by one of skill in the art from the Reactor Systems configurations disclosed.

Structured Catalyst Bed Turning now to the structured catalyst bed, similar beds have been disclosed in the prior art in relation to many catalyst promoted reactions. See for example U.S. Pat. Nos. 4,731,229; 5,073,236; 5,266,546; 5,431,890; 5,730,843; USUS2002068026; US20020038066; US20020068026; US20030012711; US20060065578; US20070209966; US20090188837; US2010063334; US2010228063; US20110214979; US20120048778; US20150166908; US20150275105; 20160074824; 20170101592 and US20170226433, the contents of which are incorporated herein by reference. While some disclosures involved reactive distillations conditions, one of skill in the art will note these disclosures involve the product being distilled from heavier bottoms or feedstock materials. For example heavy and light naphtha streams are desulfurized with the desired light naphtha being the desired product for the gasoline pool and the heavy naphtha either recycled or sent to an FCC cracker for further upgrading. The process of the invention utilizes the reactive distillation process to remove undesired by-product hydrocarbons and gases produced by the catalytic reaction (i.e. ammonia and hydrogen sulfide) and the desired residual product is the bottoms stream that is catalytically treated, but not distilled. These aspects of the inventive concepts are disclosed in greater detail in the description of the Reactive Distillation Reactor below.

The structured catalyst beds as described above balance the catalyst density load, the catalyst activity load and the desired liquid space velocity through the reactor so an effective separation or distillation of purified lighter products can be produced. In contrast the present process functionally combines the functioning of a reactor with a stripper column or knock down drum. A further problem solved by the structured catalyst bed is to reduce the pressure drop through the catalyst beds and provision of sufficient contact of the Feedstock Mixture with the catalyst and mixing with the Activating Gas.

A first illustrative embodiment of the structured catalyst beds is shown in FIG. **5** and FIG. **6** in a side view. As illustrated in FIG. **5** is a catalyst retention composed of a pair of fluid permeable corrugated metal sheets (**102** and **104**), wherein the pair of the fluid permeable corrugated metal sheets are aligned so the corrugations are sinusoidal, have the same wave length and amplitude, but are out of phase and defining a catalyst rich space (**106**) and a catalyst lean space (**108**). The catalyst rich space will be loaded with one or more catalyst materials and optionally inert packing materials. The catalyst lean space (**108**) may be left empty or it may be loaded with inert packing such as ceramic beads, inactive (non-metal containing) catalyst support, glass beads, rings, wire or plastic balls and the like. These inert packing materials may serve the role of assisting in the mixing of the Activating Gas with the Feedstock MFO, facilitate the removal or separation of gaseous by products (i.e. hydrogen sulfide or ammonia) from the process mixture or facilitate the separation of any hydrocarbon by-products.

FIG. **6** shows in side perspective a plurality of catalyst retention structures (**110**, **112** and **114**) formed into a structured catalyst bed (**116**). Structural supports (**118**) may be optionally incorporated into the structured catalyst bed to lend rigidity as needed. As shown the catalyst rich spaces are radially aligned so the catalyst rich spaces of one catalyst retention structure is aligned with the catalyst rich structure of the adjacent layers. In the illustrated configuration, the radial angle between adjacent layers is 0° (or 180°). One of skill in the art will appreciate that the angle of radial alignment between adjacent layers may be varied from 0° to 180°, preferably between 20° and 160° and more preferably 90° so the catalyst rich areas in one layer are perpendicular to the adjacent layers. It will be further appreciated that the alignment of a particular set of three or more layers need not be the same. A first layer may be aligned along and define the 0° axis relative to the other two layers; a second adjacent layer may be radially aligned along a 450 angle relative to the first layer; and the third layer aligned along a 900 angle relative to the first layer. This pattern of alignment may be continued until the desired number of layers is achieved. It also should be appreciated that it may be desirable to angle of the catalyst rich spaces (ie. the plane of the catalyst retention structure), relative to the flow of Feedstock HMFO and Activating Gas within the reaction vessel. This relative angle is referred to herein as the inclination angle. As shown in FIG. **6**, the inclination angel is perpendicular (90°) to the flow of Feedstock HMFO and Activating Gas through the reactor vessel. However, it will be appreciated that the inclination level may be varied between 0°, in which case the catalyst rich spaces are vertically aligned with the side walls of the reactor vessel and 900 in which case the catalyst rich spaces are perpendicular to the side walls of the reactor vessel. By varying both the radial alignment and the inclination angle of the catalyst rich spaces, one can achieve a wide variety and be able to optimize the flow of Feedstock HMFO though the reactor vessel with minimal plugging/coking.

A second illustrative embodiment of the structured catalyst beds is shown in FIG. **7** and FIG. **8** in a side view. As illustrated in FIG. **7**, catalyst retention structure (**200**) comprises a flat fluid permeable metal sheet (**202**) and a corrugated fluid permeable metal sheet (**204**) aligned to be co-planar and defining a catalyst rich space (**206**) and a catalyst lean space (**208**). As with the prior illustrative embodiment, the catalyst rich space will contain one or more catalyst materials and optionally inert packing materials and the catalyst lean space will be empty or optionally contain inert packing materials. FIG. **8** shows in side perspective a plurality of catalyst retention structures (**210**, **212** and **214**) formed into a structured catalyst bed (**216**). Structural supports (**218**) may be optionally incorporated into the structured catalyst bed to lend rigidity as needed. As shown the catalyst rich spaces are radially aligned so the catalyst rich spaces of one catalyst retention structure is perpendicular with the catalyst rich structure of the adjacent layers. In the illustrated configuration, the radial angle between adjacent layers is 90°. The same considerations of radial alignment and inclination of the catalyst retention structures described above will apply to this embodiment. The principle benefit of the illustrated structured catalyst bed is that the manufacturing process because affixing the flat fluid permeable sheet and the corrugated fluid permeable sheet will be greatly simplified. Further as illustrated, if the corrugated sheet is constructed using 90° angle corrugations, each catalyst retention structure can withstand much greater weight loadings than if the corrugations are sinusoidal.

The loading of the catalyst structures will depend upon the particle size of the catalyst materials and the activity level of the catalyst. The structures should be loaded so the open space will be at least 10 volume % of the overall structural volume, and preferably will be up to about 65% of the overall structural volume. Active catalyst materials should be loaded in the catalyst support structure at a level dependent upon the catalyst activity level and the desired level of treatment. For example a catalyst material highly active for desulfurization may be loaded at a lower density than a less active desulfurization catalyst material and yet still achieve the same overall balance of catalyst activity per volume. One of skill in the art will appreciate that by systematically varying the catalyst loaded per volume and the catalyst activity level one may optimize the activity level and fluid permeability levels of the structured catalyst bed. In one such example, the catalyst density is such that over 50% of the open space in the catalyst rich space, which may occupy only have of the over space within the structured catalyst bed. In another example catalyst rich space is fully loaded (i.e. dense packed into each catalyst rich space), however the catalyst rich space may occupy only 30 volume % of the overall structured catalyst bed. It will be appreciated that the catalyst density in the catalyst rich space may vary between 30 vol % and 100 vol % of the catalyst rich space. It will be further appreciated that that catalyst rich space may occupy as little as 10 vol % of the overall structured catalyst bed or it may occupy as much as 80 vol % of the overall structured catalyst bed.

The liquid hourly space velocity within the structured catalyst beds should be between 0.05 oil/hour/m^3 catalyst and 10.0 oil/hour/m^3 catalyst; preferably between 0.08 oil/hour/m^3 catalyst and 5.0 oil/hour/m^3 catalyst and more preferably between 0.1 oil/hour/m^3 catalyst and 3.0 oil/hour/m^3 catalyst to achieve deep desulfurization using a highly active desulfurization catalyst and this will achieve a product with sulfur levels below 0.1 ppmw. However, it will be appreciated by one of skill in the art that when there is lower catalyst density, it may be desirable to adjust the space velocity to value outside of the values disclosed.

Reactive Distillation Reactor As in FIG. **9**, a reactive Reactor System as contemplated by the present invention will comprise a reactor vessel (**400**) within which one or more structured beds are provided (**402**, **404** and **406**). One of skill in the art will note that heated Feedstock Mixture (D') enters the reactor vessel in the upper portion of the reactor via line (**9***b*) above the structured catalyst beds (**402**, **404** and **406**). When elements are the same as those previously disclosed, the same reference number is utilized for continuity within the disclosure. Entry of the heated Feedstock Mixture (D') above the structured catalyst beds (**402**, **404** and **406**) may be facilitated by a distribution tray or similar device not shown. It will also be noted that each of the structured catalyst beds is different in appears, the reason for this will now described. The upper most structured catalyst bed (**402**) will be preferably loaded with a demetallization catalyst and in a structure optimized for the demetallization of the Feedstock mixture. The middle structured catalyst bed (**404**) will preferably be loaded with a transition catalyst material or a mixture of demetallization and low activity desulfurization catalyst. The lower most structured catalyst bed will be preferably loaded with a desulfurization catalyst of moderate to high activity. A gas sparger or separation tray (**408**) is below structured catalyst tray **406**. In this way, the Feedstock Mixture flows from the upper portion of the reactor to the lower portion of the reactor and will be transformed into Reaction Mixture (E) which exits the bottom of the reactor via line **11***a*.

As shown, make up Activating Gas C''' will be provided via line **414** to both quench and create within the reactor a counter-current flow of Activating gas within the reactor. One of skill in the art will appreciate this flow may also be connected to the reactor so make up Activating gas is also injected between structured catalyst beds **406** and **404** and **404** and **402**. In the upper portion of the reactor, inert distillation packing beds (**410** and **412**) may be located. It may be desirably and optionally it is preferable for the lower most upper beds (**410**) to be a structured catalyst bed as well with catalyst for the desulfurization of the distillate materials. In such an instance a liquid draw tray, or a downcomer tray or similar liquid diversion tray (**414**) is inserted so a flow of middle to heavy distillate (G') can be removed from the upper portion of the reactor via line (**426**). Light hydrocarbons (i.e. lighter than middle distillate exits the top of the reactor via line (**416**) and passes through heat exchanger (**7**) to help with heat recovery. This stream is then directed to the reflux drum (**418**) in which liquids are collected for use as reflux materials. The reflux loop to the upper reactor is completed via reflux pump (**422**) and reflux line (**424**). That portion of the lights not utilized in the reflux are combined with similar flows (F and G) via lines **13***a* and **13***b* respectively.

One of skill in the art of reactor design will note that unlike the prior art reactive distillation processes and reactor designs, the present invention presents multiple novel and non-obvious (i.e. inventive step) features. One such aspect the Feedstock Mixture enters the upper portion of the reactor above the structured catalytic beds. In doing so it is transformed into Reaction Mixture that exits the bottom of the reactor. One of skill in the art will appreciate that by this flow, the majority of HMFO material (which is characterized as being residual in nature) will not be volatile or distilled, but any byproduct gases, distillate hydrocarbons or light hydrocarbons are volatilized into the upper portion of the reactor. The reactor will be hydraulically designed so the majority of the volume of the liquid components having residual properties in the Feedstock Mixture will exit the lower portion of the reactor, preferably over 75% vol. of the volume of the liquid components having residual properties in the Feedstock Mixture will exit the lower portion of the reactor and even more preferably over 90% vol. of the volume of the liquid components having residual properties in the Feedstock Mixture will exit the lower portion of the reactor. This is in contract with the prior art where the majority of the desired products exit the upper portion of the reactor via distillation and the residual bottoms portions are recycled or sent to another refinery unit for further processing.

In a variation of the above illustrative embodiment, one or more fixed bed reactor(s) containing, solid particle filtering media such as inactive catalyst support, inert packing materials, selective absorption materials such as sulfur absorption media, demetallization catalyst or combinations and mixtures of these may be located upstream of the Reactive distillation reactor. In one alternative embodiment a reactor with a mixture of hydrodemetallization catalyst, decarbonization catalysts and inert materials will act as a guard bed, protecting the Reactive distillation reactor from high levels of metals, concarbon (CCR) and other contaminants that can reduce the run length of the Reactive distillation reactor. In another embodiment, the upstream reactors are loaded within inert packing materials and deactivated catalyst to remove solids followed by a reactor loaded within hydrodemetallization catalyst. One of skill in the art will appreciate these upstream reactors may allow the upstream reactors to be taken out of service and catalysts changed out without shutting down or affecting operation of the Reactive distillation reactor.

In another variation of the above illustrative embodiment in FIG. **9**, a fired reboiler can be added to the lower portion of the reactive distillation reactor. Such a configuration would take a portion of the Reaction Mixture from the bottom of the reactor prior to its exit via line **11***a*, pass it through a pump and optionally a heater, and reintroduce the material into the reactor above tray **408** and preferably above the lowermost structured catalyst bed (**406**). The purpose of the reboiler will be to add or remove heat within the reactive distillation reactor, and will increase column traffic, because of this reboiler loop a temperature profile in the reactive distillation reactor will be controlled and more distillate product(s) may be taken. We assume severity in the column could be increased to increase the hydrocracking activity within the reactive distillation reactor increasing the distillate production. Because of the washing effect caused by refluxing Reaction Mixture back into the reactive distillation reactor, coking and fouling of catalysts should be minimized, allowing for extending run lengths.

Catalyst in Reactor System: Generally a reactor vessel in each Reactor System is loaded with one or more process catalysts, preferably within one or more structured beds. The selection of the catalyst system in each reactor vessel is a function of feedstock properties, product requirements and operating constraints and optimization of the process catalyst for the specific reactor and can be carried out by routine trial and error by one of ordinary skill in the art.

Within the scope of the present invention, the process catalyst(s) (also called catalyst materials) comprise at least one metal selected from the group consisting of the metals each belonging to the groups 6, 8, 9 and 10 of the Periodic Table, and more preferably a mixed transition metal catalyst such as Ni—Mo, Co—Mo, Ni—W or Ni—Co—Mo are utilized. The metal is preferably supported on a porous inorganic oxide catalyst carrier. The porous inorganic oxide catalyst carrier is at least one carrier selected from the group consisting of alumina, alumina/boria carrier, a carrier containing metal-containing aluminosilicate, alumina/phosphorus carrier, alumina/alkaline earth metal compound carrier, alumina/titania carrier and alumina/zirconia carrier. The preferred porous inorganic oxide catalyst carrier is alumina. The pore size and metal loadings on the carrier may be systematically varied and tested with the desired feedstock and process conditions to optimize the properties of the Product HMFO. One of skill in the art knows that demetallization using a transition metal catalyst (such a CoMo or NiMo) is favored by catalysts with a relatively large surface pore diameter and desulfurization is favored by supports having a relatively small pore diameter. Generally the surface area for the catalyst material ranges from 200-300 m^2/g. The systematic adjustment of pore size and surface area, and transition metal loadings activities to preferentially form a demetallization catalyst or a desulfurization catalyst are well known and routine to one of skill in the art.

The catalyst selection utilized within and for loading the Reactor System may be preferentially designed with a catalyst loading scheme that results in the Feedstock Mixture first contacting a catalyst bed that with a catalyst preferential to demetallization followed downstream by a bed of catalyst with mixed activity for demetallization and desulfurization followed downstream by a catalyst bed with high desulfurization activity. A bed of inert materials may be loaded as the first bed followed by a bed with high demetallization activity the two beds acting as a guard bed to minimize metal deposits on the subsequent the desulfurization beds.

The objective of the Reactor System is to treat the Feedstock HMFO at the severity required to meet the Product HMFO specification. Demetallization, denitrogenation and hydrocarbon hydrogenation reactions may also occur to some extent when the process conditions are optimized so the performance of the Reactor System achieves the required level of desulfurization. Hydrocracking is preferably minimized to reduce the volume of hydrocarbons formed as by-product hydrocarbons to the process. The objective of the process is to selectively remove the environmental contaminates from Feedstock HMFO and minimize the formation of unnecessary by-product hydrocarbons (C_1-C_8 hydrocarbons).

The process conditions in each reactor vessel will depend upon the feedstock, the catalyst utilized and the desired properties of the Product HMFO. Variations in conditions are to be expected by one of ordinary skill in the art and these may be determined by pilot plant testing and systematic optimization of the process. With this in mind it has been found that the operating pressure, the indicated operating temperature, the ratio of the Activating Gas to Feedstock HMFO, the partial pressure of hydrogen in the Activating Gas and the space velocity all are important parameters to consider. The operating pressure of the Reactor System should be in the range of 250 psig and 3000 psig, preferably between 1000 psig and 2500 psig and more preferably between 1500 psig and 2200 psig. The indicated operating temperature of the Reactor System should be 500° F. to 900°

F., preferably between 650° F. and 850° F. and more preferably between 680° F. and 800° F. The ratio of the quantity of the Activating Gas to the quantity of Feedstock HMFO should be in the range of 250 scf gas/bbl of Feedstock HMFO to 10,000 scf gas/bbl of Feedstock HMFO, preferably between 2000 scf gas/bbl of Feedstock HMFO to 5000 scf gas/bbl of Feedstock HMFO and more preferably between 2500 scf gas/bbl of Feedstock HMFO to 4500 scf gas/bbl of Feedstock HMFO. The Activating Gas should be selected from mixtures of nitrogen, hydrogen, carbon dioxide, gaseous water, and methane, so Activating Gas has an ideal gas partial pressure of hydrogen (pH2) greater than 80% of the total pressure of the Activating Gas mixture (P) and preferably wherein the Activating Gas has an ideal gas partial pressure of hydrogen (pH2) greater than 95% of the total pressure of the Activating Gas mixture (P). The Activating Gas may have a hydrogen mole fraction in the range between 80% of the total moles of Activating Gas mixture and more preferably wherein the Activating Gas has a hydrogen mole fraction between 80% and 99% of the total moles of Activating Gas mixture. The liquid hourly space velocity within the Reactor System should be between 0.05 oil/hour/m^3 catalyst and 1.0 oil/hour/m^3 catalyst; preferably between 0.08 oil/hour/m^3 catalyst and 0.5 oil/hour/m^3 catalyst and more preferably between 0.1 oil/hour/m^3 catalyst and 0.3 oil/hour/m^3 catalyst to achieve deep desulfurization with product sulfur levels below 0.1 ppmw.

The hydraulic capacity rate of the Reactor System should be between 100 bbl of Feedstock HMFO/day and 100,000 bbl of Feedstock HMFO/day, preferably between 1000 bbl of Feedstock HMFO/day and 60,000 bbl of Feedstock HMFO/day, more preferably between 5,000 bbl of Feedstock HMFO/day and 45,000 bbl of Feedstock HMFO/day, and even more preferably between 10,000 bbl of Feedstock HMFO/day and 30,000 bbl of Feedstock HMFO/day. The desired hydraulic capacity may be achieved in a single reactor vessel Reactor System or in a multiple reactor vessel Reactor System as described.

Oil Product Stripper System Description: The Oil Product Stripper System (**19**) comprises a stripper column (also known as a distillation column or exchange column) and ancillary equipment including internal elements and utilities required to remove hydrogen, hydrogen sulfide and light hydrocarbons lighter than diesel from the Product HMFO. Such systems are well known to one of skill in the art, see U.S. Pat. Nos. 6,640,161; 5,709,780; 5,755,933; 4,186,159; 3,314,879 U.S. Pat. Nos. 3,844,898; 4,681,661; or U.S. Pat. No. 3,619,377 the contents of which are incorporated herein by reference, a generalized functional description is provided herein. Liquid from the Hot Separator (**13**) and Cold Separator (**7**) feed the Oil Product Stripper Column (**19**). Stripping of hydrogen and hydrogen sulfide and light hydrocarbons lighter than diesel may be achieved via a reboiler, live steam or other stripping medium. The Oil Product Stripper System (**19**) may be designed with an overhead system comprising an overhead condenser, reflux drum and reflux pump or it may be designed without an overhead system. The conditions of the Oil Product Stripper may be optimized to control the bulk properties of the Product HMFO, more specifically viscosity and density. It is also contemplated that a second draw (not shown) may be included to withdraw a distillate product, preferably a middle to heavy distillate.

Amine Absorber System Description: The Amine Absorber System (**21**) comprises a gas liquid contacting column and ancillary equipment and utilities required to remove sour gas (i.e. hydrogen sulfide) from the Cold Separator vapor feed so the resulting scrubbed gas can be recycled and used as Activating Gas. Because such systems are well known to one of skill in the art, see U.S. Pat. Nos. 4,425,317; 4,085,199; 4,080,424; 4,001,386; which are incorporated herein by reference, a generalized functional description is provided herein. Vapors from the Cold Separator (**17**) feed the contacting column/system (**19**). Lean Amine (or other suitable sour gas stripping fluids or systems) provided from OSBL is utilized to scrub the Cold Separator vapor so hydrogen sulfide is effectively removed. The Amine Absorber System (**19**) may be designed with a gas drying system to remove the any water vapor entrained into the Recycle Activating Gas (C'). The absorbed hydrogen sulfide is processed using conventional means OSBL in a tail gas treating unit, such as a Claus combustion sulfur recovery unit or sulfur recovery system that generates sulfuric acid.

These examples will provide one skilled in the art with a more specific illustrative embodiment for conducting the process disclosed and claimed herein:

Example 1

Overview: The purpose of a pilot test run is to demonstrate that feedstock HMFO can be processed through a reactor loaded with commercially available catalysts at specified conditions to remove environmental contaminates, specifically sulfur and other contaminants from the HMFO to produce a product HMFO that is MARPOL compliant, that is production of a Low Sulfur Heavy Marine Fuel Oil (LS-HMFO) or Ultra-Low Sulfur Heavy Marine Fuel Oil (USL-HMFO).

Pilot Unit Set Up: The pilot unit will be set up with two 434 cm^3 reactors arranged in series to process the feedstock HMFO. The lead reactor will be loaded with a blend of a commercially available hydrodemetallization (HDM) catalyst and a commercially available hydro-transition (HDT) catalyst. One of skill in the art will appreciate that the HDT catalyst layer may be formed and optimized using a mixture of HDM and HDS catalysts combined with an inert material to achieve the desired intermediate/transition activity levels. The second reactor will be loaded with a blend of the commercially available hydro-transition (HDT) and a commercially available hydrodesulfurization (HDS). One can load the second reactor simply with a commercially hydrodesulfurization (HDS) catalyst. One of skill in the art will appreciate that the specific feed properties of the Feedstock HMFO may affect the proportion of HDM, HDT and HDS catalysts in the reactor system. A systematic process of testing different combinations with the same feed will yield the optimized catalyst combination for any feedstock and reaction conditions. For this example, the first reactor will be loaded with ⅔ hydrodemetallization catalyst and ⅓ hydro-transition catalyst. The second reactor will be loaded with all hydrodesulfurization catalyst. The catalysts in each reactor will be mixed with glass beads (approximately 50% by volume) to improve liquid distribution and better control reactor temperature. For this pilot test run, one should use these commercially available catalysts: HDM: Albemarle KFR 20 series or equivalent; HDT: Albemarle KFR 30 series or equivalent; HDS: Albemarle KFR 50 or KFR 70 or equivalent. Once set up of the pilot unit is complete, the catalyst can be activated by sulfiding the catalyst using dimethyldisulfide (DMDS) in a manner well known to one of skill in the art.

Pilot Unit Operation: Upon completion of the activating step, the pilot unit will be ready to receive the feedstock HMFO and Activating Gas feed. For the present example, the Activating Gas can be technical grade or better hydrogen gas. The mixed Feedstock HMFO and Activating Gas will be provided to the pilot plant at rates and operating conditions as specified: Oil Feed Rate: 108.5 ml/h (space velocity=0.25/h); Hydrogen/Oil Ratio: 570 Nm3/m3 (3200 scf/bbl); Reactor Temperature: 372° C. (702° F.); Reactor Outlet Pressure:13.8 MPa(g) (2000 psig).

One of skill in the art will know that the rates and conditions may be systematically adjusted and optimized depending upon feed properties to achieve the desired product requirements. The unit will be brought to a steady state for each condition and full samples taken so analytical tests can be completed. Material balance for each condition should be closed before moving to the next condition.

Expected impacts on the Feedstock HMFO properties are: Sulfur Content (wt %): Reduced by at least 80%; Metals Content (wt %): Reduced by at least 80%; MCR/Asphaltene Content (wt %): Reduced by at least 30%; Nitrogen Content (wt %): Reduced by at least 20%; C1-Naphtha Yield (wt %): Not over 3.0% and preferably not over 1.0%.

Process conditions in the Pilot Unit can be systematically adjusted as per Table 1 to assess the impact of process conditions and optimize the performance of the process for the specific catalyst and feedstock HMFO utilized.

TABLE 1

Optimization of Process Conditions

Case	HC Feed Rate (ml/h), [LHSV(/h)]	Nm^3 H_2/ m^3 oil/ scf H_2/bbl oil	Temp (° C./° F.)	Pressure (MPa(g)/ psig
Baseline	108.5 [0.25]	570/3200	372/702	13.8/2000
T1	108.5 [0.25]	570/3200	362/684	13.8/2000
T2	108.5 [0.25]	570/3200	382/720	13.8/2000
L1	130.2 [0.30]	570/3200	372/702	13.8/2000
L2	86.8 [0.20]	570/3200	372/702	13.8/2000
H1	108.5 [0.25]	500/2810	372/702	13.8/2000
H2	108.5 [0.25]	640/3590	372/702	13.8/2000
S1	65.1 [0.15]	620/3480	385/725	15.2/2200

In this way, the conditions of the pilot unit can be optimized to achieve less than 0.5% wt. sulfur product HMFO and preferably a 0.1% wt. sulfur product HMFO. Conditions for producing ULS-HMFO (i.e. 0.1% wt. sulfur product HMFO) will be: Feedstock HMFO Feed Rate: 65.1 ml/h (space velocity=0.15/h); Hydrogen/Oil Ratio: 620 Nm^3/m^3 (3480 scf/bbl); Reactor Temperature: 385° C. (725° F.); Reactor Outlet Pressure: 15 MPa(g) (2200 psig)

Table 2 summarizes the anticipated impacts on key properties of HMFO.

TABLE 2

Expected Impact of Process on Key Properties of HMFO

Property	Minimum	Typical	Maximum
Sulfur Conversion/Removal	80%	90%	98%
Metals Conversion/Removal	80%	90%	100%
MCR Reduction	30%	50%	70%
Asphaltene Reduction	30%	50%	70%
Nitrogen Conversion	10%	30%	70%
C1 through Naphtha Yield	0.5%	1.0%	4.0%
Hydrogen Consumption (scf/bbl)	500	750	1500

Table 3 lists analytical tests to be carried out for the characterization of the Feedstock HMFO and Product HMFO. The analytical tests include those required by ISO for the Feedstock HMFO and the product HMFO to qualify and trade in commerce as ISO compliant residual marine fuels. The additional parameters are provided so that one skilled in the art can understand and appreciate the effectiveness of the inventive process.

TABLE 3

Analytical Tests and Testing Procedures

Sulfur Content	ISO 8754 or ISO 14596 or ASTM D4294
Density @ 15° C.	ISO 3675 or ISO 12185
Kinematic Viscosity @ 50° C.	ISO 3104
Pour Point, ° C.	ISO 3016
Flash Point, ° C.	ISO 2719
CCAI	ISO 8217, ANNEX B
Ash Content	ISO 6245
Total Sediment-Aged	ISO 10307-2
Micro Carbon Residue, mass %	ISO 10370
H2S, mg/kg	IP 570
Acid Number	ASTM D664
Water	ISO 3733
Specific Contaminants	IP 501 or IP 470 (unless indicated otherwise)
Vanadium	or ISO 14597
Sodium	
Aluminum	or ISO 10478
Silicon	or ISO 10478
Calcium	or IP 500
Zinc	or IP 500
Phosphorous	IP 500
Nickle	
Iron	
Distillation	ASTM D7169
C:H Ratio	ASTM D3178
SARA Analysis	ASTM D2007
Asphaltenes, wt %	ASTM D6560
Total Nitrogen	ASTM D5762
Vent Gas Component Analysis	FID Gas Chromatography or comparable

Table 4 contains the Feedstock HMFO analytical test results and the Product HMFO analytical test results expected from the inventive process that indicate the production of a LS HMFO. It will be noted by one of skill in the art that under the conditions, the levels of hydrocarbon cracking will be minimized to levels substantially lower than 10%, more preferably less than 5% and even more preferably less than 100 of the total mass balance.

TABLE 4

Analytical Results

	Feedstock HMFO	Product HMFO
Sulfur Content, mass %	3.0	0.3
Density @ 15° C., kg/m^3	990	950 (1)
Kinematic Viscosity @ 50° C., mm^2/s	380	100 (1)
Pour Point, ° C.	20	10
Flash Point, ° C.	110	100 (1)
CCAI	850	820
Ash Content, mass %	0.1	0.0
Total Sediment-Aged, mass %	0.1	0.0
Micro Carbon Residue, mass %	13.0	6.5
H2S, mg/kg	0	0
Acid Number, mg KO/g	1	0.5
Water, vol %	0.5	0
Specific Contaminants, mg/kg		
Vanadium	180	20
Sodium	30	1
Aluminum	10	1
Silicon	30	3
Calcium	15	1
Zinc	7	1
Phosphorous	2	0
Nickle	40	5
Iron	20	2

TABLE 4-continued

Analytical Results	Feedstock HMFO	Product HMFO
Distillation, ° C./° F.		
IBP	160/320	120/248
5% wt	235/455	225/437
10% wt	290/554	270/518
30% wt	410/770	370/698
50% wt	540/1004	470/878
70% wt	650/1202	580/1076
90% wt	735/1355	660/1220
FBP	820/1508	730/1346
C:H Ratio (ASTM D3178)	1.2	1.3
SARA Analysis		
Saturates	16	22
Aromatics	50	50
Resins	28	25
Asphaltenes	6	3
Asphaltenes, wt %	6.0	2.5
Total Nitrogen, mg/kg	4000	3000

Note:
(1) property will be adjusted to a higher value by post process removal of light material via distillation or stripping from product HMFO.

The product HMFO produced by the inventive process will reach ULS HMFO limits (i.e. 0.1% wt. sulfur product HMFO) by systematic variation of the process parameters, for example by a lower space velocity or by using a Feedstock HMFO with a lower initial sulfur content.

Example 2: RMG-380 HMFO

Pilot Unit Set Up: A pilot unit was set up as noted above in Example 1 with these changes: the first reactor was loaded with: as the first (upper) layer encountered by the feedstock 70% vol Albemarle KFR 20 series hydrodemetallization catalyst and 30% vol Albemarle KFR 30 series hydrotransition catalyst as the second (lower) layer. The second reactor was loaded with 20% Albemarle KFR 30 series hydrotransition catalyst as the first (upper) layer and 80% vol hydrodesulfurization catalyst as the second (lower) layer. The catalyst was activated by sulfiding the catalyst with dimethyldisulfide (DMDS) in a manner well known to one of skill in the art.

Pilot Unit Operation: Upon completion of the activating step, the pilot unit was ready to receive the feedstock HMFO and Activating Gas feed. The Activating Gas was technical grade or better hydrogen gas. The Feedstock HMFO was a commercially available and merchantable ISO 8217 (2017) compliant HMFO, except for a high sulfur content (2.9 wt %). The mixed Feedstock HMFO and Activating Gas was provided to the pilot plant at rates and conditions as specified in Table 5 below. The conditions were varied to optimize the level of sulfur in the product HMFO material.

TABLE 5

Case	HC Feed Rate (ml/h), [LHSV(/h)]	Nm^3 H_2/ m^3 oil/ scf H_2/bbl oil	Temp (° C./° F.)	Pressure (MPa(g)/ psig	HMFO Sulfur % wt.
Baseline	108.5 [0.25]	570/3200	371/700	13.8/2000	0.24
T1	108.5 [0.25]	570/3200	362/684	13.8/2000	0.53
T2	108.5 [0.25]	570/3200	382/720	13.8/2000	0.15
L1	130.2 [0.30]	570/3200	372/702	13.8/2000	0.53
S1	65.1 [0.15]	620/3480	385/725	15.2/2200	0.10

TABLE 5-continued

Case	HC Feed Rate (ml/h), [LHSV(/h)]	Nm^3 H_2/ m^3 oil/ scf H_2/bbl oil	Temp (° C./° F.)	Pressure (MPa(g)/ psig	HMFO Sulfur % wt.
P1	108.5 [0.25]	570/3200	371/700	/1700	0.56
T2/P1	108.5 [0.25]	570/3200	382/720	/1700	0.46

Analytical data for a representative sample of the feedstock HMFO and representative samples of product HMFO are below:

TABLE 6

Analytical Results-HMFO (RMG-380)	Feedstock	Product	Product
Sulfur Content, mass %	2.9	0.3	0.1
Density @ 15° C., kg/m^3	988	932	927
Kinematic Viscosity @ 50° C., mm^2/s	382	74	47
Pour Point, ° C.	−3	−12	−30
Flash Point, ° C.	116	96	90
CCAI	850	812	814
Ash Content, mass %	0.05	0.0	0.0
Total Sediment-Aged, mass %	0.04	0.0	0.0
Micro Carbon Residue, mass %	11.5	3.3	4.1
H2S, mg/kg	0.6	0	0
Acid Number, mg KO/g	0.3	0.1	>0.05
Water, vol %	0	0.0	0.0
Specific Contaminants, mg/kg			
Vanadium	138	15	<1
Sodium	25	5	2
Aluminum	21	2	<1
Silicon	16	3	1
Calcium	6	2	<1
Zinc	5	<1	<1
Phosphorous	<1	2	1
Nickle	33	23	2
Iron	24	8	1
Distillation, ° C./° F.			
IBP	178/352	168/334	161/322
5% wt	258/496	235/455	230/446
10% wt	298/569	270/518	264/507
30% wt	395/743	360/680	351/664
50% wt	517/962	461/862	439/822
70% wt	633/1172	572/1062	552/1026
90% wt	>720/>1328	694/1281	679/1254
FBP	>720/>1328	>720/>1328	>720/>1328
C:H Ratio (ASTM D3178)	1.2	1.3	1.3
SARA Analysis			
Saturates	25.2	28.4	29.4
Aromatics	50.2	61.0	62.7
Resins	18.6	6.0	5.8
Asphaltenes	6.0	4.6	2.1
Asphaltenes, wt %	6.0	4.6	2.1
Total Nitrogen, mg/kg	3300	1700	1600

In Table 6, both feedstock HMFO and product HMO exhibited observed bulk properties consistent with ISO 8217 (2017) for a merchantable residual marine fuel oil, except that the sulfur content of the product HMO was reduced as noted above when compared to the feedstock HMO.

One of skill in the art will appreciate that the above product HMO produced by the inventive process has achieved not only an ISO 8217 (2017) compliant LS HMFO (i.e. 0.5% wt. sulfur) but also an ISO 8217 (2017) compliant ULS HMFO limits (i.e. 0.10% wt. sulfur) product HMFO.

Example 3: RMK-500 HMFO

The feedstock to the pilot reactor utilized in example 2 above was changed to a commercially available and merchantable ISO 8217 (2017) RMK-500 compliant HMFO, except that it has high environmental contaminates (i.e. sulfur (3.3 wt %)). Other bulk characteristic of the RMI-500 feedstock high sulfur HMFO are provide below:

TABLE 7

Analytical Results-Feedstock HMFO (RMK-500)	
Sulfur Content, mass %	3.3
Density @ 15° C., kg/m^3	1006
Kinematic Viscosity @ 50° C., mm^2/s	500

The mixed Feedstock (RMK-500) HMFO and Activating Gas was provided to the pilot plant at rates and conditions and the resulting sulfur levels achieved in the table below

TABLE 8

	Process Conditions				Product (RMK-500) sulfur % wt.
Case	HC Feed Rate (ml/h), [LHSV(/h)]	Nm^3 H_2/m^3 oil/ scf H_2/bbl oil	Temp (° C./° F.)	Pressure (MPa(g)/ psig	
A	108.5 [0.25]	640/3600	377/710	13.8/2000	0.57
B	95.5 [0.22]	640/3600	390/735	13.8/2000	0.41
C	95.5 [0.22]	640/3600	390/735	11.7/1700	0.44
D	95.5 [0.22]	640/3600	393/740	10.3/1500	0.61
E	95.5 [0.22]	640/3600	393/740	17.2/2500	0.37
F	95.5 [0.22]	640/3600	393/740	8.3/1200	0.70
G	95.5 [0.22]	640/3600	416/780	8.3/1200	

The resulting product (RMK-500) HMO exhibited observed bulk properties consistent with the feedstock (RMK-500) HMO, except that the sulfur content was reduced as noted in the above table.

One of skill in the art will appreciate that the above product HMFO produced by the inventive process has achieved a LS MFO (i.e. 0.5% wt. sulfur) product HMFO having bulk characteristics of a ISO 8217 (2017) compliant RMK-500 residual fuel oil. It will also be appreciated that the process can be successfully carried out under non-hydrocracking conditions (i.e. lower temperature and pressure) that substantially reduce the hydrocracking of the feedstock material. When conditions were increased to much higher pressure (Example E) a product with a lower sulfur content was achieved, however some observed that there was an increase in light hydrocarbons and wild naphtha production.

It will be appreciated by those skilled in the art that changes could be made to the illustrative embodiments described above without departing from the broad inventive concepts thereof. It is understood, therefore, that the inventive concepts disclosed are not limited to the illustrative embodiments or examples disclosed, but it should cover modifications within the scope of the inventive concepts as defined by the claims.

The invention claimed is:

1. A process for the production of a Product Heavy Marine Fuel Oil, the process comprising: mixing a quantity of Feedstock Heavy Marine Fuel Oil, wherein Feedstock Heavy Marine Fuel Oil complies with Table 2 of ISO 8217 (2017) as a residual marine fuel and has a sulfur content (ISO 14596 or ISO 8754) greater than 0.5% wt., with a quantity of Activating Gas mixture to give a Feedstock Mixture; contacting the Feedstock Mixture with one or more hydrotreating catalysts under reactive distillation conditions in a reaction vessel, wherein the reactive distillation conditions promote the formation of a Process Mixture from said Feedstock Mixture, wherein said one or more catalysts are in the form of a structured catalyst bed, and wherein a majority by weight of the Process Mixture is comprised of a Product Heavy Marine Fuel Oil component and a minority by weight of the Process Mixture is comprised of bulk gaseous component and by-product hydrocarbon component; and wherein the reactive distillation conditions promote separating the Product Heavy Marine Fuel Oil portion of the Process Mixture from the bulk gaseous components of the Process Mixture; and the by-product hydrocarbon components from the Product Heavy Marine Fuel Oil; and discharging the Product Heavy Marine Fuel Oil from the bottom of the reaction vessel.

2. The process of claim **1**, wherein the structured catalyst bed comprises a plurality of catalyst retention structures, each catalyst retention structure composed of a pair of fluid permeable corrugated metal sheets, wherein the pair of the fluid permeable corrugated metal sheets are aligned such that the corrugations are out of phase and thereby defining a catalyst rich space and a catalyst lean space within the structured catalyst bed, wherein within the catalyst rich space there is one or more catalyst materials and optionally inert packing materials.

3. The process of claim **1**, wherein the structured catalyst bed contains one or more catalysts to promote the transformation of the Feedstock Mixture to a Process Mixture wherein the structured catalyst bed comprises a plurality of catalyst retention structures, each catalyst retention structure composed of a flat fluid permeable metal sheet and a corrugated fluid permeable metal, wherein the flat fluid permeable corrugated metal and the corrugated fluid permeable sheet are aligned to be co-planer and thereby defining a catalyst rich space and a catalyst lean space within the structured catalyst bed, wherein within the catalyst rich space there is one or more catalyst materials and optionally inert packing materials and the catalyst lean space is empty or optionally contains inert packing materials.

4. The process of claim **1** wherein the Product Heavy Marine Fuel Oil complies with Table 2 of ISO 8217 (2017) residual marine fuel and has a sulfur content (ISO 14596 or ISO 8754) between the range of 0.50 mass % to 0.05 mass %.

5. The process of claim **4**, wherein said Feedstock Heavy Marine Fuel Oil has: a combination of kinematic viscosity at 50° C. (ISO 3104) and a density at 15° C. (ISO 3675) to give a CCAI in the range of 780 to 870; a flash point (ISO 2719) no lower than 60° C.; a total sediment—aged (ISO 10307-2) of less than 0.10 mass %; a carbon residue—micro method (ISO 10370) less than 20 mass %; and an aluminum plus silicon (ISO 10478) content less than 60 mg/kg.

6. The process of claim **4**, wherein said Product Heavy Marine Fuel Oil has: a kinematic viscosity at 50° C. (ISO 3104); a density at 15° C. (ISO 3675) to give a CCAI in the range of 780 to 870; a flash point (ISO 2719) no lower than 60° C.; a total sediment—aged (ISO 10307-2) of less than 0.10 mass %; a carbon residue—micro method (ISO 10370) less than 20 mass %; and an aluminum plus silicon (ISO 10478) content less than 60 mg/kg.

7. The process of claim **1** wherein the catalyst materials comprises: a porous inorganic oxide catalyst carrier and a transition metal catalyst, wherein the porous inorganic oxide catalyst carrier is at least one carrier selected from the group consisting of alumina, alumina/boria carrier, a carrier containing metal-containing aluminosilicate, alumina/phosphorus carrier, alumina/alkaline earth metal compound carrier, alumina/titania carrier and alumina/zirconia carrier, and wherein the transition metal catalyst is one or more metals selected from the group consisting of group 6, 8, 9 and 10 of the Periodic Table; and wherein the Activating Gas is selected from mixtures of nitrogen, hydrogen, carbon dioxide, gaseous water, and methane, such that Activating Gas has an ideal gas partial pressure of hydrogen (p_{H2}) greater than 80% of the total pressure of the Activating Gas mixture (P).

8. The process of claim **7**, wherein the reactive distillation conditions comprise: the ratio of the quantity of the Activating Gas to the quantity of Feedstock Heavy Marine Fuel Oil is in the range of 250 scf gas/bbl of Feedstock Heavy Marine Fuel Oil to 10,000 scf gas/bbl of Feedstock Heavy Marine Fuel Oil; a the total pressure is between of 250 psig and 3000 psig; and, the indicated temperature is between of 500° F. to 900° F., and, wherein the liquid hourly space velocity is between 0.05 oil/hour/m^3 catalyst and 1.0 oil/hour/m^3 catalyst.

9. The process of claim **7**, wherein the structured catalyst bed comprises a plurality of catalyst retention structures, each catalyst retention structure composed of a pair of fluid permeable corrugated metal sheets, wherein the pair of the fluid permeable corrugated metal sheets are aligned such that the corrugations are out of phase and thereby defining a catalyst rich space and a catalyst lean space within the structured catalyst bed, wherein within the catalyst rich space there is one or more catalyst materials and optionally inert packing materials.

10. The process of claim **7**, wherein the structured catalyst bed comprises a plurality of catalyst retention structures, each catalyst retention structure composed of a flat fluid permeable metal sheet and a corrugated fluid permeable metal, wherein the flat fluid permeable corrugated metal and the corrugated fluid permeable sheet are aligned to be co-planer and thereby defining a catalyst rich space and a catalyst lean space within the structured catalyst bed, wherein within the catalyst rich space there is one or more catalyst materials and optionally inert packing materials and the catalyst lean space is empty or optionally contains inert packing materials.

11. The process of claim **7** wherein the Product Heavy Marine Fuel Oil complies with Table 2 of ISO 8217 (2017) residual marine fuel and has a sulfur content (ISO 14596 or ISO 8754) between the range of 0.50 mass % to 0.05 mass %.

12. The process of claim **11**, wherein said Feedstock Heavy Marine Fuel Oil has: a combination of kinematic viscosity at 50° C. (ISO 3104) and a density at 15° C. (ISO 3675) to give a CCAI in the range of 780 to 870; a flash point (ISO 2719) no lower than 60° C.; a total sediment—aged (ISO 10307-2) of less than 0.10 mass %; a carbon residue—micro method (ISO 10370) less than 20 mass %; and an aluminum plus silicon (ISO 10478) content less than 60 mg/kg.

13. The process of claim **12**, wherein said Product Heavy Marine Fuel Oil has: a kinematic viscosity at 50° C. (ISO 3104) and a density at 15° C. (ISO 3675) to give a CCAI in the range of 780 to 870; a flash point (ISO 2719) no lower than 60° C.; a total sediment—aged (ISO 10307-2) of less than 0.10 mass %; a carbon residue—micro method (ISO 10370) less than 20 mass %; and an aluminum plus silicon (ISO 10478) content less than 60 mg/kg.

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